What We Will Cover Today

- Spotlight reflections remaining and clarification on points
  - Attend exactly 4 to get 3% added to your final grade.
- Semester Project Meetings
  - Clarification of general expectations for the presentations
  - Presentation tips
- Probability Quiz Review
  - And some new questions!! (maybe)
- Lecture Review
  - Quick review to ensure understanding of the material covered by Dr. Helfand.

Science Spotlights

Semester Project Presentations !

Group Presentations

- 20 minutes
  - 15-minute presentation + 5-minutes for questions
  - Each group member must speak
  - Delegation of duties is up to the group
  - You will be graded as a group and an individual :
    - 85 points for the overall group presentation
    - 10 points for individual contribution (Presentation & Ability to answer questions)
    - 5 points for 3 questions

How The Grade Will Be Determined

- 85 points for the overall group presentation
  - Content
    - Do you cover all necessary subtopics/questions
  - Clarity of Slides
    - Don’t over stuff the slides;
      - No more than 8 sentences in a single slide
      - No more than 4 figures or tables on a single slide)
    - Slides should build
  - Presentation Flow
    - The slide order should make intuitive sense OR include a “Road map” to guide the audience
    - Transitions between points should be smooth
    - Try to avoid filler words

How you will be graded

- 10 points for individual contribution
  - Each presenter should speak clearly and loudly
  - Description of slides should be smooth
  - Try not to read off slides or note cards too much
  - Answer reasonable questions satisfactorily

How you will be graded

- 5 points for 3 questions
- Group questions are reasonable and thoughtful

Slides

- There is no limit on the number of slides, but remember the presentation is only 15 minutes long!
  - 6 people = 2.5 minutes per person
- Slide deck must be submitted via email by **6pm** the day before the presentation. (12/02)

Slides

- On the first slide:
  - Title for the presentation
  - Names of all contributing members
  - Semester : Fall 2025

Slides

- The topics covered should reflect your research papers:
  - Research question
  - Introduction/lit review
  - Explanation of methods and data analysis strategy
  - Discussion of Results
  - Discussion of project limitations and future research projects/directions
- Include graphics and illustrations to make it more interesting
  - Use pictures
  - Add labels and arrows to the images to direct the audience’s attention

Introduction/ Research Question/

- Articulates a clear, reasonable research question given existing theory and research, All constructs and variables are appropriately defined.

Explanation of methods: Participants and Procedure

- The description of the procedure and sample is comprehensive
- Includes the overall number of participants
- A breakdown by key demographic characteristics
- Information about how the sample was obtained (survey etc.)
- Using a flow chart can help relay the methods well

Explanation of methods: Data Analysis

- Analytical methods are sufficiently specific
- Mention controls (don’t add a lot of detail on controls)
- Describe test groups or hypothesis related results and figures
- Don’t add charts, add specific relevant data points.

Results

- All results are presented correctly with correct statistical terminology
- Clear, easily interpretable
- Tables/figures are presented well
- Only use plots and graphs that you need to explain.

Discussion: results

- A succinct and logical interpretation of results is present.
- The literature presented in the introduction is revisited with an eye towards understanding the results.

Discussion: Limitations & Future Directions

- There is a discussion of possible confounds or alternative explanations, as well as a clear indication of limitations and future potential studies

Group Opinion

- Group opinions on the findings or techniques
  - Are the findings believable ?
  - Did you identify any additional limitations
  - How you would improve the studies
  - How would you build on these studies
  - Was the data interesting, were the findings surprising, etc.)

  - Pick 1 or 2 of these topics

Questions

- Wait until the end of the presentation to ask questions
- Each group aside from the presenting group will ask a question
  - Every group will need to come up with 3 questions total (1 for each group)
  - Come up with reasonable questions, the idea is to learn from your classmates not to stump them.
- I will ask additional questions until all group members have a chance to answer 1question.

General Presentation Tips

Presentation Tip: What to say and how to say it

- Communicate the key ideas
  - Emphasize the important parts
  - Skip things that are standard, obvious, or too complicated
- Don’t get bogged down in details
  - Too many details are out of place in a presentation
  - Give a sufficiently detailed overview.
- Structure the talk
  - Inform the audience of where you are and where you are headed
  - The slides should be broken down into several distinct parts (Intro, method, result, discussion)(Intro, technology, Practical applications, conclusions)
  - Steer the audience from one part to the next.

Presentation Tip: What to say and how to say it

- Know your audience
  - Make sure the talk is prepared at the right level
  - No one in the room will be an expert.
- Use an organized approach
  - In addition to your slides having structure the presentation also needs structure
  - Introduction (Research Question, lit review, define key terms)
  - Body (Hypothesis, Method, results, result interpretation)
  - Conclusion(Limitations of the study, future directions, opinions)

Presentation Tip: The audience

- Practice the talk
  - Verbal, not just mental
  - Even minimal practice helps tremendously
- Use Repetition
  - Into: Tell them what you are going to tell them
  - Body: Tell them
  - Conclusion: Tell them what you told them
- Don't Over-run
  - Keep an eye on the time
  - Practice with the time limit
- Try to maintain eye contact with the audience
- Try Not to Get Anxious
  - Practice and Experience

4 team presentation formats

- Hosted Format
- Relay Format
- Hybrid Format
- Popcorn Format

Hosted Format

- Relies on one constant speaker (the host)
  - Opens and concludes the presentation and facilitates transitions between speakers
  - Deliver the introduction and conclusion
  - Transitions include linking of previous group members points to the next group members points
  - Host should be a strong speaker !
- Pros
  - Smooth transitions between speakers
  - Keeps the presentation engaging
- Cons
  - Relies on one person to execute the most crucial moments
  - A lot of responsibility on one team member
  - A 'bad' host can tank a presentation
  - The host can overtake the presentation

Relay Format

- Each presenter has an equal part
- Three players
  - An opener (Intro mostly)
  - Main point speaker(s) (Methods & Results mostly)
  - A closer (Discussion mostly)
- Each speaker facilitates their own handoff to the next speaker
- Pros
  - Most common and natural format
  - Doesn't emphasize one speaker over the rest of the group
  - Un-hosted transitions can be faster
- Cons
  - It may be hard to actually evenly divide the content, some group members will be responsible for more than others
  - No one makes a unique impact and each group member's presentation skills are on full display
  - Increased chance of mistakes between transitions

Hybrid Format

- Uses the same speaker as the opener (introduction) and closer(conclusion)
- Un-hosted transitions
- Pros
  - Stabilizes the introduction and conclusion
  - Reduces the burden on one speaker
  - Allows the best speaker to help improve the presentation with their skill without taking it over.
- Cons
  - One person is still very important since they are responsible for both the opening and closing, If they mess up the presentation will be noticeably affected
  - Un-hosted transitions mean the possibility for miscues increases

Popcorn Format

- Everyone helps with every part of the presentation
  - Scripted conversations between members or improvised
- No hand-offs between speakers since everyone can speak
- Rehearsal is really important
- Anyone can pop in at any time (popcorn)
- Pros
  - Highly engaging for the audience
  - Involves all team members equally
- Cons
  - Requires extensive rehearsal
  - It can seem chaotic and the speakers can seem unprepared
  - Team members could interrupt or contradict each other if improvised

As a presenter

- Make sure the content is understandable for your crowd.
  - Provide an explanation for terms that are not easily understood by mentioning them.
- Prepare and rehearse ahead of time !
- Delegate roles and practice alone AND together

As an audience member

- Pay attention to the presentations
  - No looking at phones or laptops
  - No napping
  - No distracting behavior
  - No eating
  - Save all questions for the end of the presentation
  - Raise your had and wait for acknowledgement before asking a question.
- Ask questions about the presentation

Probability

Again!

Probability of getting A AND B, where A and B are independent events: P(A)P(B)

Probability of drawing all four Queens from a deck of cards (without replacement, you keep each card that you draw).

$$[P(A)]^{\text{1st card}} \times [P(A)]^{\text{2nd card}} \times [P(A)]^{\text{3rd card}} \times [P(A)]^{\text{4th card}} =$$

$$= (4/52) \times (3/51) \times (2/50) \times (1/49) = 3.6 \times 10^{-6}$$

Probability of getting NEITHER A NOR B, where A and B are independent events:

Probability of neither A nor B happening is:

$$P(\text{not } A)\, P(\text{not } B) = [1 – P(A)]\, [1 – P(B)]$$

- What is the probability that a card chosen randomly from a 52-deck of cards will be NOT a Queen NOR a heart?

P(not Queen not heart) = P(not Queen)* P(not heart);

P(not Queen) = 1 – P(Queen); P(Queen)= 4/52

P(not heart) = 1 – P(heart); P(heart) = 13/52

$$= [1 – (4/52)]\, [1 – (13/52)] =$$

$$= (48/52)(39/52) = 36/52 = 0.69$$

Probability of getting A <u>OR</u> B, where A and B are independent events:

For two independent events A and B there are two possible mutually exclusive outcomes:

- <u>Either A or B happens (only A happens, or only B happens, or both A and B happen).</u>
- Neither A, nor B happens.

Remember: the sum of the individual probabilities for all possible mutually exclusive outcomes of an event is 1.

Probability of getting A OR B, where A and B are independent events:

- Probability of neither A nor B happening is:

$$P(\text{not } A)\, P(\text{not } B) = [1 - P(A)]\,[1 - P(B)]$$

- Probability of A or B happening is:

$$1 - P(\text{not } A)\, P(\text{not } B) = 1 - [1 - P(A)]\,[1 - P(B)]$$

- Probability of A or B or C or D happening is:

$$1 - [1 - P(A)]\,[1 - P(B)]\,[1 - P(C)]\,[1 - P(D)]$$

Probability of event A occurring <u>at least once</u> in N tries:

- Probability of A not happening in a single try:

$$P(\text{not } A) = [1 - P(A)]$$

- Probability of A not happening in all N tries:

$$P(\text{not A in 1}^{st}\text{ try})\ P(\text{not A in 2}^{nd}\text{ try})\ \ldots\ (\text{P not A in N}^{th}\text{ try})$$

$$= [1 - P(A)]^N$$

- Probability of A happening at least once in N tries:

$$1 - [1 - P(A)]^N$$

Probability of event A occurring at least once in N tries: $1 - [1 - P(A)]^N$

- The recurrence interval for a meteor of size X is 2×10^8 years. What is the probability that something that size will hit Earth at least once in your lifetime? Assume you will live 80 years.

$$P(A) = P(\text{hitting the Earth in a given year}) = 1/(2 \times 10^8)$$

$$P(\text{at least once in 80 years}) = 1 - [1 - P(A)]^N$$

$$= 1 - [\,1 - (1/2 \times 10^8)]^{80} = 4 \times 10^{-7}$$

Probability of event A occurring at least once in N tries: $1 - [1 - P(A)]^N$

- What is the probability that something of size X will hit Earth at least once in the next 6 months?

$P(A) = P(\text{hitting the Earth in a given month}) = 1/(2 \times 10^8 \times 12)$

$= 4.2 \times 10^{-10}$

$P(\text{at least once in 6 month}) = 1 - [1 - P(A)]^N$

$= 1 - [\,1 - (4.2 \times 10^{-10})]^6 = 2.5 \times 10^{-9}$

Probability of event A occurring <u>at least once</u> in N tries: $1 - [1 - P(A)]^N$

- What is the probability that at least one person in our class is infected with COVID-19 (Assume 300,000 infection out of 330 million US pop)?

$$P(A) = P(\text{one person being infected}) = (3 \times 10^5)/(330 \times 10^6)$$

$$= 9.09 \times 10^{-4}$$

$$P(\text{at least one infection in 20 students}) = 1 - [\,1 - (9.09 \times 10^{-4})]^{20}$$

$$= 0.018$$

Probability Practice

<u>Question 1</u>

You have 20 pairs of socks in your drawer. Out of those, you have 5 identical pairs of yellow socks. Note that you didn’t fold your socks, so you have just stuffed single socks in your drawer without matching them. **You pick up two socks from the drawer without looking. What is the probability that you picked a pair of yellow socks?**

<u>Question 1</u>

You have 20 pairs of socks in your drawer. Out of those, you have 5 identical pairs of yellow socks. Note that you didn't fold your socks, so you have just stuffed single socks in your drawer without matching them.

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1^{st} sock AND 2^{nd} sock have to be yellow

$P(1^{st}\text{ yellow}) \times P(2^{nd}\text{ yellow}) = (10/40) \times (9/39) = 0.058 = 5.8\%$

Question 2

A jelly bean jar contains 30 blue, 20 green, 40 yellow, and 10 red jelly beans.

i) If you pick one jelly bean from the jar what are the odds of picking a red one?

ii) If you pick one jelly bean from the jar what are the odds of not picking a green one?

iii) If you pick five jelly beans from the jar (without replacement) what are the odds of not picking a green one? What are the odds that at least one of them is green?

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i) $10/(30+20+40+10) = 0.1$

ii) $1 - P(\text{picking one green}) = 1 - (20/100) = 0.8$

iii) $P(\text{not green the } 1^{st} \text{ time}) \times P(\text{not green } 2^{nd} \text{ time}) \ldots =$

$(1\text{-}20/100) \times (1\text{-}20/99) \times (1\text{-}20/98) \times (1\text{-}20/97) \times (1\text{-}20/96) =$

$0.8 \times 0.798 \times 0.796 \times 0.794 \times 0.792 = 0.32$

$P\ (\text{at least one green in 5}) = 1 - P(\text{not green in 5}) = 1 - 0.32 = 0.68$

Question 3

(i) Given the weather forecast shown below, calculate the probability that it will rain EVERY day next week. Assume that rain events on any given day are independent of rain events on any other day. **(ii) Given the weather forecast shown below, calculate the probability that it will rain at least one day in the next week.** Assume that rain events on any given day are independent of rain events on any other day.

	Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
							
Chance of rain	10%	20%	30%	30%	20%	5%	5%

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Chance of rain	10%	20%	30%	30%	20%	5%	5%

(i) P(rain every day) = 0.1 x 0.2 x 0.3 x 0.3 x 0.2 x 0.05 x 0.05 = $9\text{x}10^{-7}$- tiny probability, about 1 in a million.

(ii) P(rain at least one day) = 1-(0.9 x 0.8 x 0.7 x 0.7 x 0.8 x 0.95 x 0.95) = 0.75 (fairly large probability - 75%)

Question 4

A carrier of a genetic disease has one mutated allele (out of two alleles), but does not display the disease phenotype. **(i)** An Ashkenazy Jewish (AJ) couple, both carriers of the Cystic Fibrosis (CF) allele, ask their doctor what the probability is that they will have a child with CF. **What is the doctor's answer?** Note that for CF, two copies of the mutated allele are required in order to have the disease. **(ii) What would be the probability that another AJ couple's child develops both Tay-Sachs Disease (TSD) and CF, if the mother is a carrier for CF and the father is a carrier for TSD?** Note that for TSD, just like for CF, two copies of the mutated allele are required in order to have the disease.

Disease	General population carrier frequency	AJ carrier frequency	Probability of affected fetus if parents pos/neg[a] in AJ	Probability of affected fetus if parents neg/neg[b] in AJ
Cystic Fibrosis (CF)	1/200	1/23	1/1,472	$1/5.4x10^5$
Tay-Sachs Disease (TSD)	1/300	1/27	1/5,204	$1/6.8x10^6$

[a]If/when genetic screening finds that one parent is a carrier and one parent is not.
[b]If/when genetic screening finds that neither parent is a carrier.

Table 4. *Carrier frequency and probability of Cystic Fibrosis (CF) and Tay-Sachs Disease (TSD) in AJ population, given specific outcomes of genetic testing. Table adapted from Scott et al. Human Mutation, 2010.*

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(i) P(CF) = P(CF gene from mom AND CF gene from dad) = ½ × ½ = 1/4.

(ii) P of a child having TSD when 1 parent is a carrier and the other is not is 1/5204.
P of a child having CF when 1 parent is a carrier and the other is not is 1/1472.
P(TSD AND CF) = (1/5204) × (1/1472) = 1/7,660,288 = 1.3×10^{-7}.

Question 5

Scientists have discovered that a change in a single nucleotide (single nucleotide polymorphism, SNP) on the MCM6 gene immediately adjacent to the lactase gene was responsible for the appearance of Lactate Persistence. Let G be the allele that leads to Lactase Persistence and let the frequency of G occurring in a given population be 70%. **(i) What is the probability any person in this population has at least one G in their genotype? (ii) What is the probability any person in this population is heterozygous in G in their genotype?**

Question 5

Scientists have discovered that a change in a single nucleotide (single nucleotide polymorphism, SNP) on the MCM6 gene immediately adjacent to the lactase gene was responsible for the appearance of Lactate Persistence. Let G be the allele that leads to Lactase Persistence and let the frequency of G occurring in a given population be 70%. **(i) What is the probability any person in this population has at least one G in their genotype? (ii) What is the probability any person in this population is heterozygous in G in their genotype?**

Given P(G) = 0.7, P(g) = 0.3,

(i) P(G from mom OR G from dad) = P(G) + P(G) – P(G)P(G) = 0.7 + 0.7 – (0.7 x 0.7) = 0.91

OR P(GG) = 0.7 x 0.7, P(Gg) = 0.7 x 0.3, P(gG) = 0.7 x 0.3 = 0.49 + 0.21 + 0.21 = 0.91

OR P(gg) = 0.3 x 0.3, P(not_gg) = 1-(0.3 x 0.3) = 0.91

(ii) P(<u>only</u> one G from mom OR <u>only</u> one G from dad) = P(G) + P(G) – P(G)P(G) – P(G)P(G) = 0.7 + 0.7 – 2(0.7 x 0.7) = 0.42;

Note: the first P(G)P(G) is the redundant double-counting, the second P(G)P(G) is the homozygous genotype.

OR P(Gg) = 0.7 x 0.3 = 0.21; P(gG) = 0.7 x 0.3 = 0.21 → P(het) = 0.21 + 0.21 = 0.42

OR P(gg) = 0.3 x 0.3, P(GG) = 0.7 x 0.7, P(not_gg,not_GG) = 1-(0.9 + 0.49)= 0.42

Homework review

1B. (i) In order to guide the CRISPR-Cas9 complex to the BRCA genes, the researchers designed a 20-base pair (bp) guide RNA. **How many possible unique 20-bp guide RNA sequences exist?**

- Possible nucleic acids for an RNA sequence = A – U – C - G

Since there are 4 possible nucleic acids at each of the 20 spacer sequence positions, there are $4^{20} = 1.1 \times 10^{12}$ *possible sequences.*

1B. (ii) A 20-bp sequence is used to guide CRISPR-Cas9 to its target site in the pancreatic cells that the researchers are aiming to edit. **What is the probability that a randomly selected site is an exact match for a given 20-bp gRNA sequence?** Assume all possible sequences are equally likely.

Assuming that all 20-bp sequences are equally likely in the human genome, the probability that any one 20-nucleotide sequence matches the spacer sequence is $P = \frac{1}{1.1x\ 10^{12}} = 9.1x10^{-13}$

Homework review

1B. (iii) In order to introduce premature STOP codons into the BRCA1 and BRCA2 genes, the researchers need to edit 4 sites in the DNA. In previous trials, the researchers found that their genome editing procedure could successfully edit a single site in a single cell 85% of the time. **Based on this information, estimate the probability that they can successfully edit all four target mutations in one pancreatic cell.**

P(one target edited) = 0.85

P(all 4 targets in one cell edited) = P(one target edited)^4 $= 0.85^4 = 0.522$ *or 52.2%*

Homework review

1B. (iv) Estimate the probability that they can successfully edit all four target mutations in every cell in a pool of 50 pancreatic cells.

P(all 50 cells edited) = P(all 4 targets in one cell edited)^{50} = $(0.85^4)^{50}$ = 7.65×10^{-15}

1B. (v) What is the probability that they can successfully edit at least one of the four target mutations in every cell in a pool of 50 pancreatic cells? [Hint: $P = 1-(1-P_{one})^n$]

Probability of one cell having at least one mutation = $1 - (1 - 0.85)^4 = 0.9995$

Probability of all 50 cells having at least one mutation = $(0.9995)^{50} = 0.975$ or 97.5%

Climate & Us

Week 1:
What We Know and
What We Don't Know

Are we seeing the impacts of climate change already?

YES! We are seeing the impacts of climate change TODAY – though these impacts are not equally felt

Climate change:
- Increasing extreme weather events
- Rising sea levels
- Changes in rainfall

Climate change is a "threat multiplier"

Climate displacement

2022: Climate-related disasters triggered more than half of new reported displacements

*Mid 2024: **90 million** of current 123 million forcibly displaced people live in countries with high-to-extreme exposure to climate-related hazards*

*By 2050: **200 million** will be in need of humanitarian assistance due to climate change*

Destruction of critical resources:
- Clean fresh water
- Agriculture
- Ocean (>3 billion rely on the ocean for food)

Sahel:
- Temperatures rising 1.5x faster than global average
- One of the world's fastest growing displacement crisis

CU1 Learning Objectives

1. Distinguish between weather and climate.
2. Explain the role that astronomical influences (sunspots and eccentricity, obliquity, and precession) play in climate.
3. Analyze the composition of the Earth's atmosphere, the recent changes that we observe, the role humans play in those change, and how we know human activity is the cause.
4. Describe the global temperature record since 1880, with a focus on the last few decades.
5. Reconstruct the Earth's energy balance by considering the flow of energy into and away from the Earth's surface and atmosphere, with a special emphasis on the role of reflectivity and greenhouse gases.

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LO1: Weather and Climate

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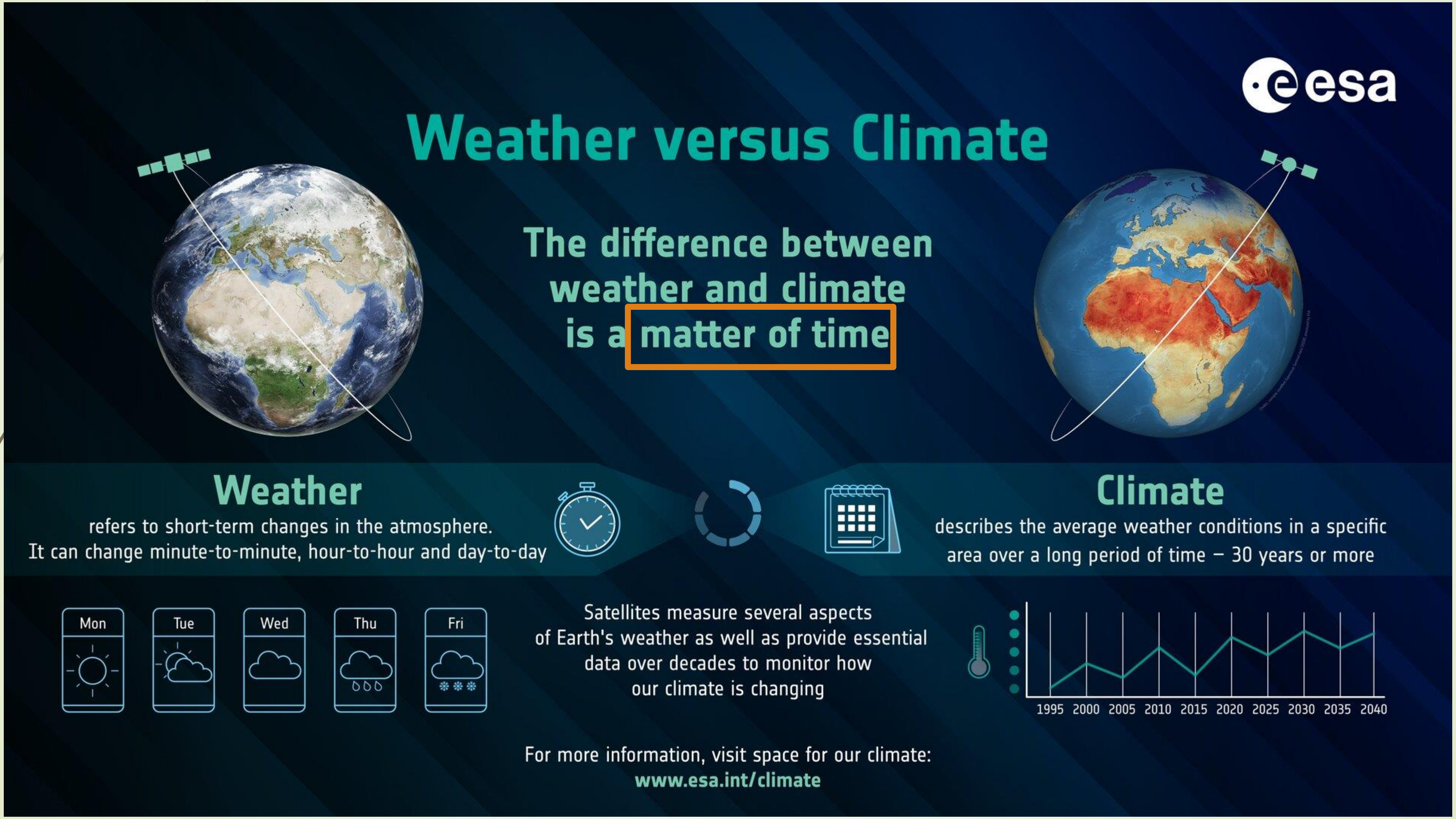
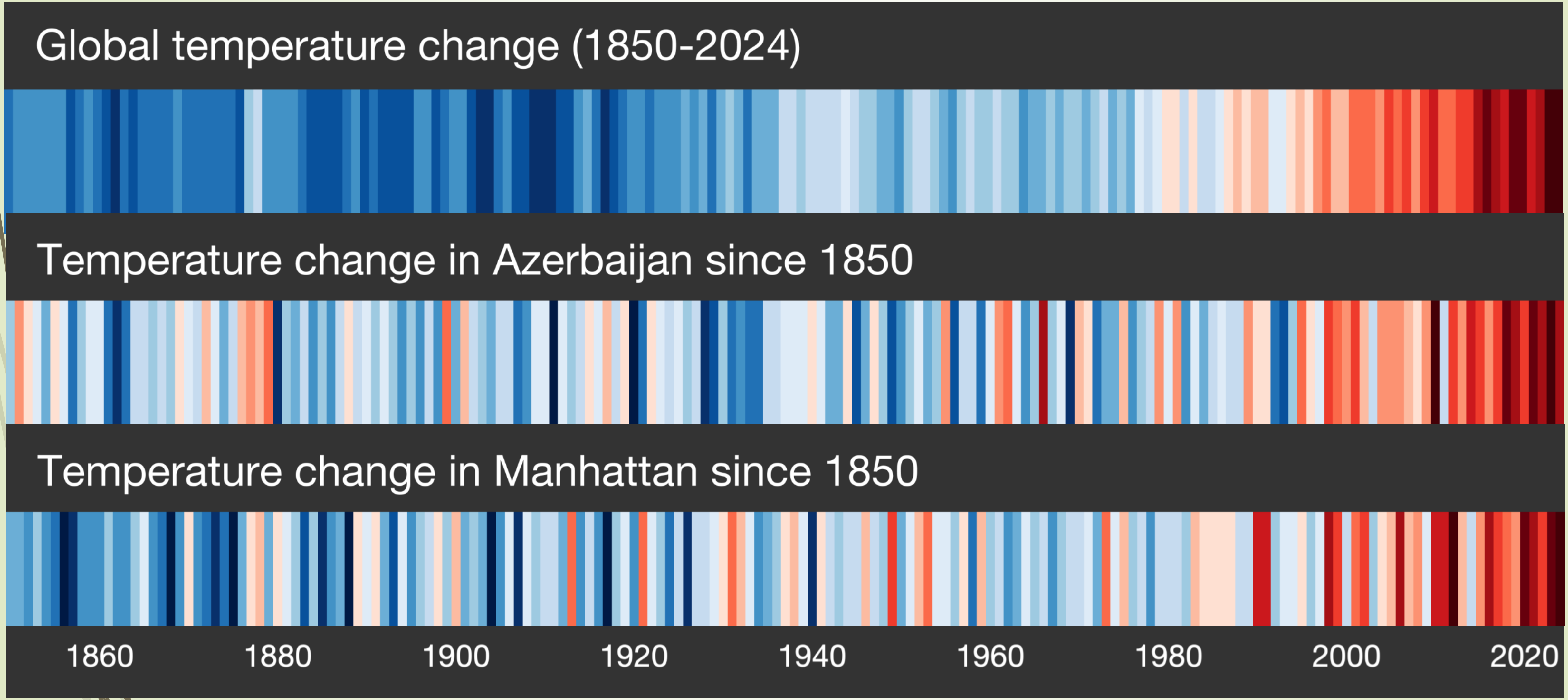


LO1: Weather and Climate

	Weather	Climate
What is the period of time?	Changes in minutes to years	Changes in decades to millennia
What are some examples?	Storms, fronts, droughts, cold snaps	Ice ages, medieval warm period
What kind of questions do we ask?	Do I take an umbrella today? Do I buy a warmer jacket this winter?	Is my home near the shore in trouble? Do I buy farmland in Canada?

LO1: Weather and Climate



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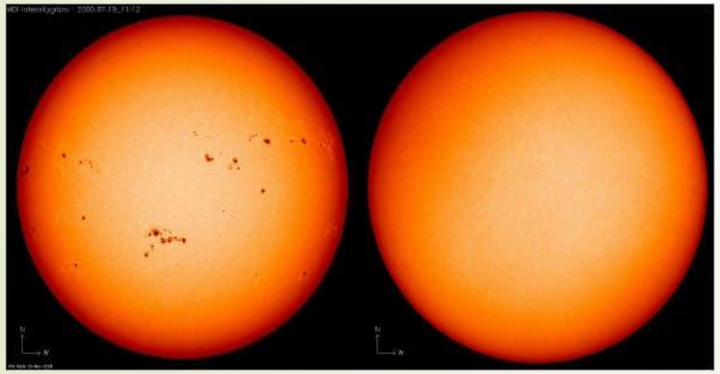
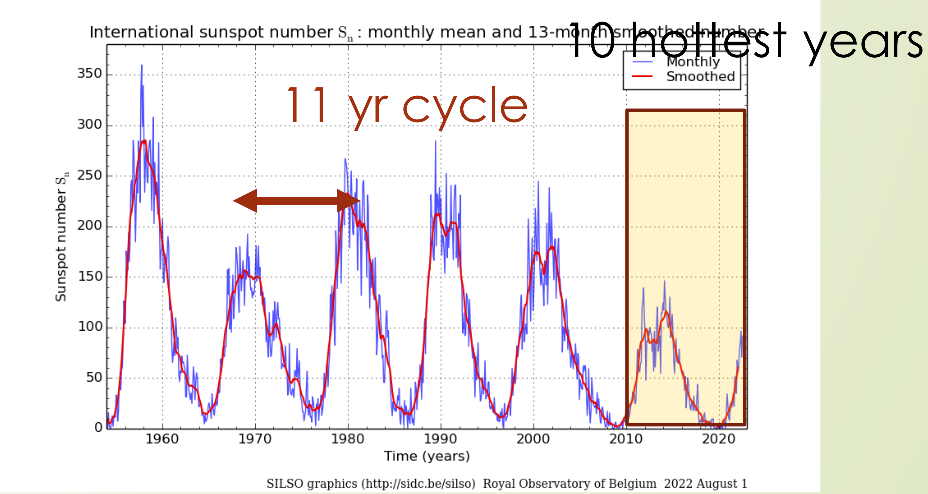
LO2: Astronomical Influences - Sunspots

Disturbances in Sun's magnetic field, which can generate energetic solar events

- Yearly avg. number of sunspots increases during periods of high solar activity
- The change in the Sun's yearly average total irradiance during an 11-year cycle is about 0.1 percent or 1.4 W/m^2
- Past decade is *lowest* solar activity recorded this century. Yet, 10 out of 10 hottest years in century during this period



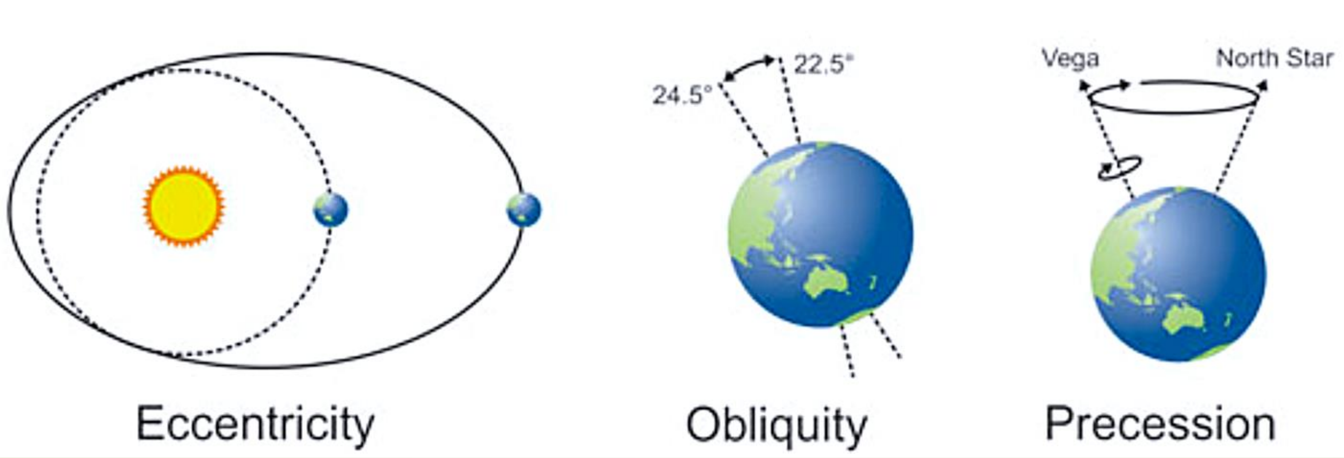
Sunspots near solar maximum on July 19, 2000, and near solar minimum on March 18, 2009 (by NASA)



LO2: Astronomical Influences - Orbital Cycles

Systematic changes in the distance and orientation of Earth relative to the Sun → affects the amount of solar radiation (insolation) Earth receives

- Affect Earth's climate over 20,000 to 100,000 yrs
- Orbital cycles (aka **Milanković Cycles**):



tal plane (41,000 yr

ed ("wobble") (23,000

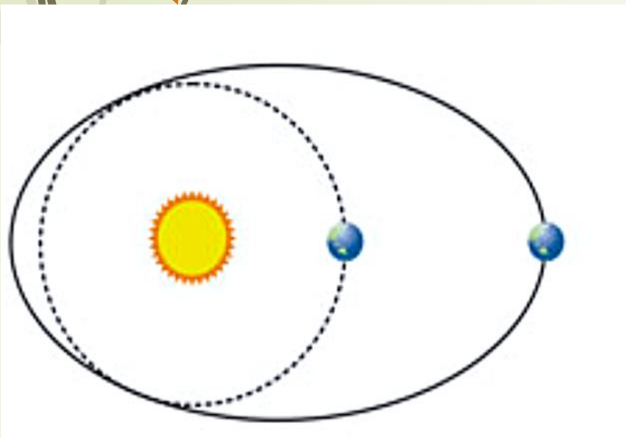
BUT the effect is NOT relevant to the recent global warming we are experiencing

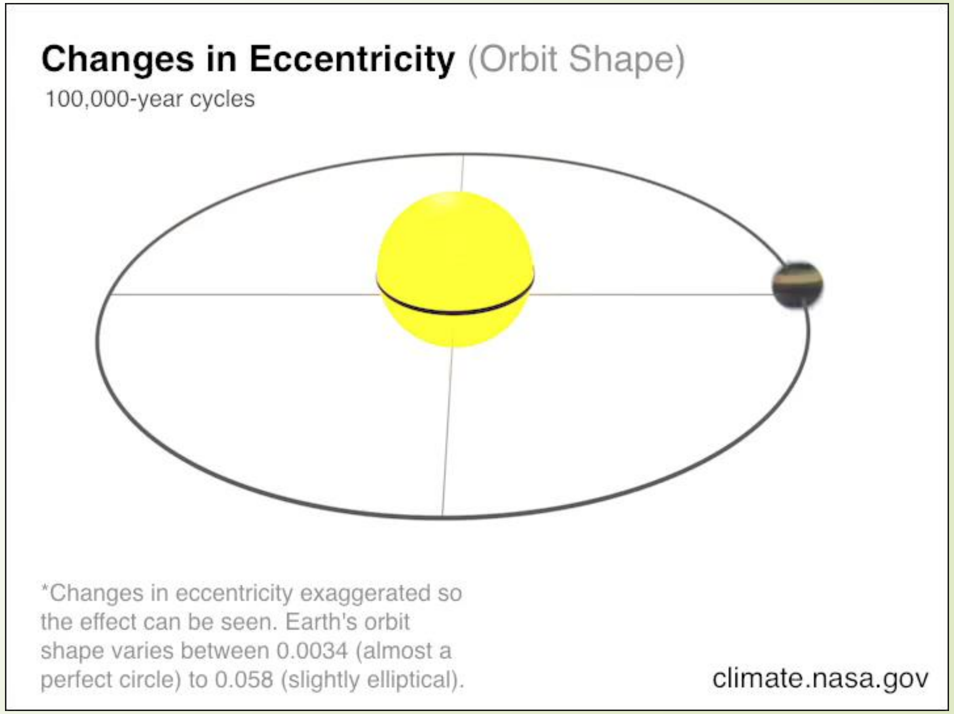
Eccentricity: 100,000 year cycle

Shape of Earth's orbit from circular to elliptical

- High eccentricity: large seasonal discrepancy in solar radiation
- Currently Earth's eccentricity is nearly circular
  - 5 million km closer to sun on Jan 3rd than on July 4th

23% more incoming solar radiation at closest approach than at farthest departure from sun



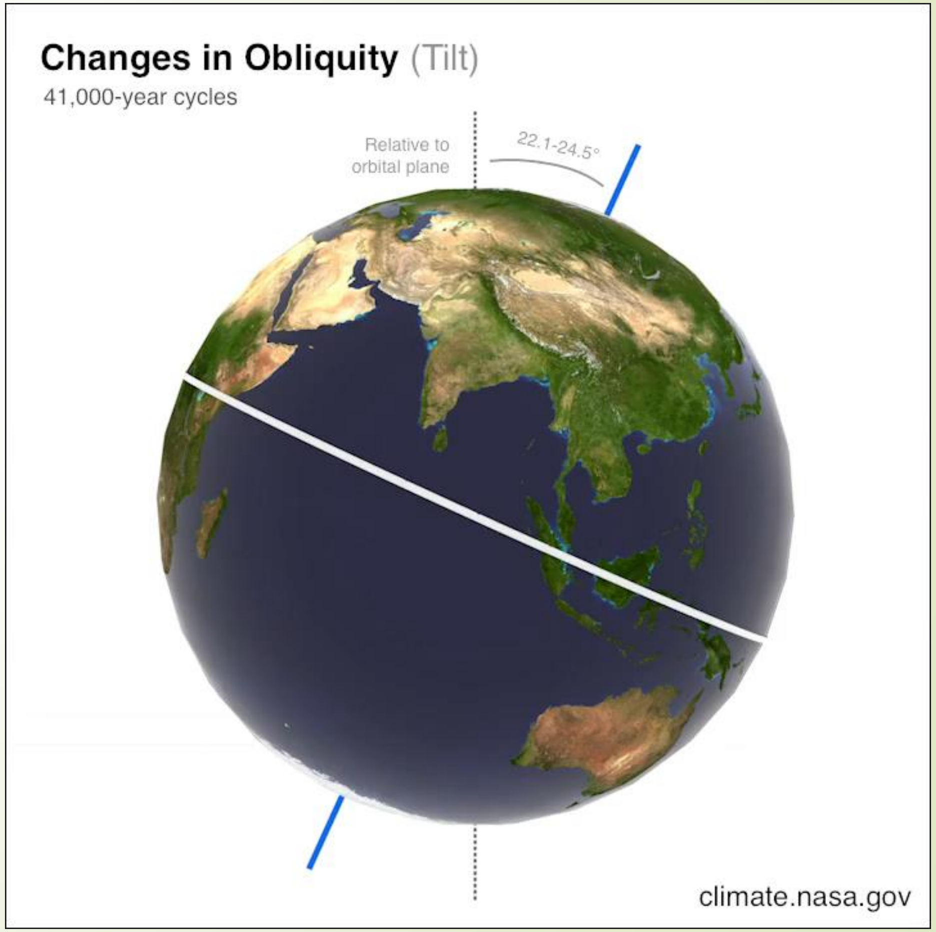


Obliquity: 41,000 year cycle

Angle of Earth's axis of rotation is tilted with respect to Earth's orbital plane

- Varies from 22.1 to 24.5 degrees (currently 23.4 degrees, ~halfway)

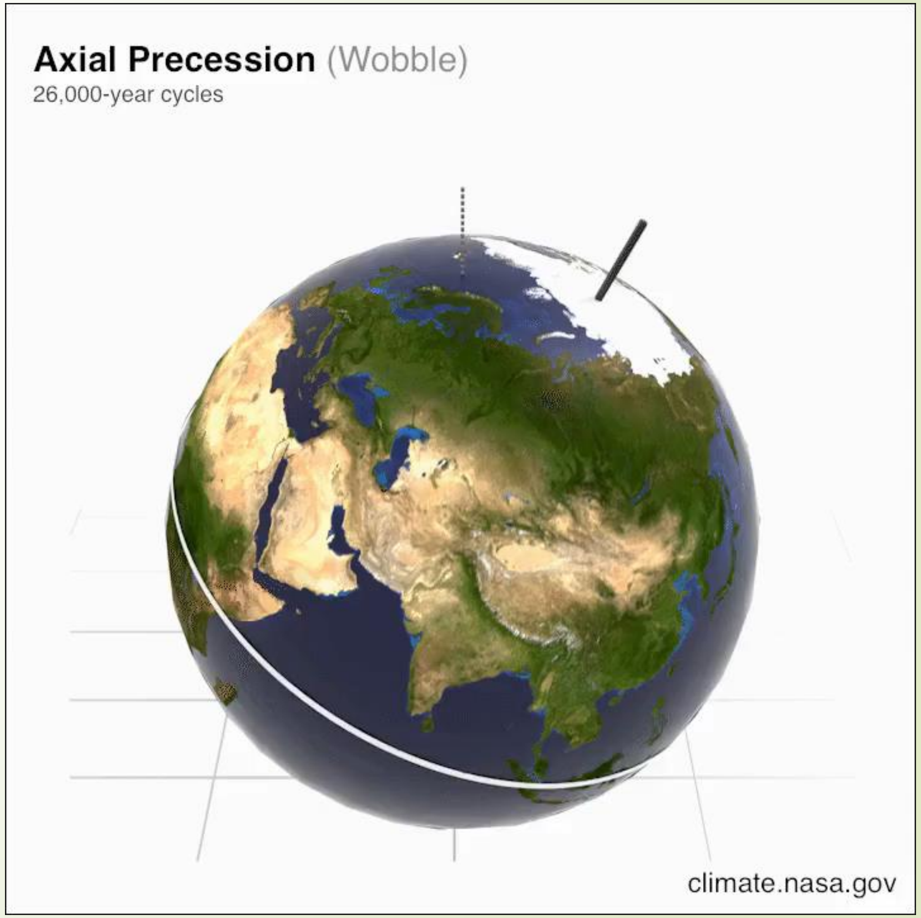
- Tilt is the reason for Earth's seasons
  - The higher the tilt angle, the more extreme our seasons are
- Effects aren't uniform globally
- As obliquity decreases, winters will be warmer and summers will be cooler



Precession: 23,000 year cycle

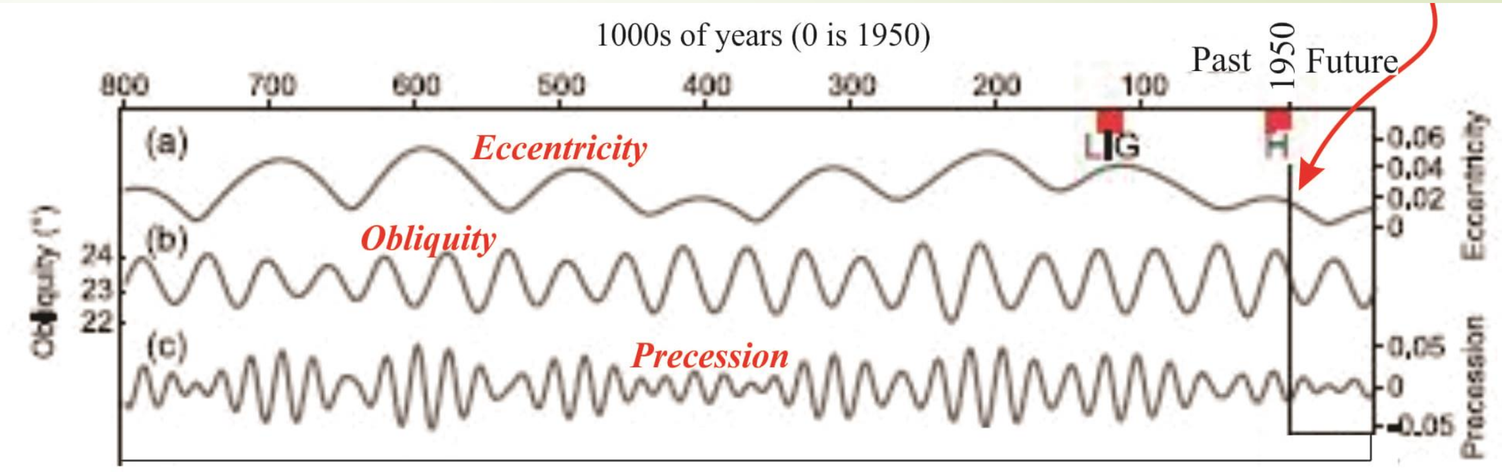
Direction Earth's axis of rotation is pointed

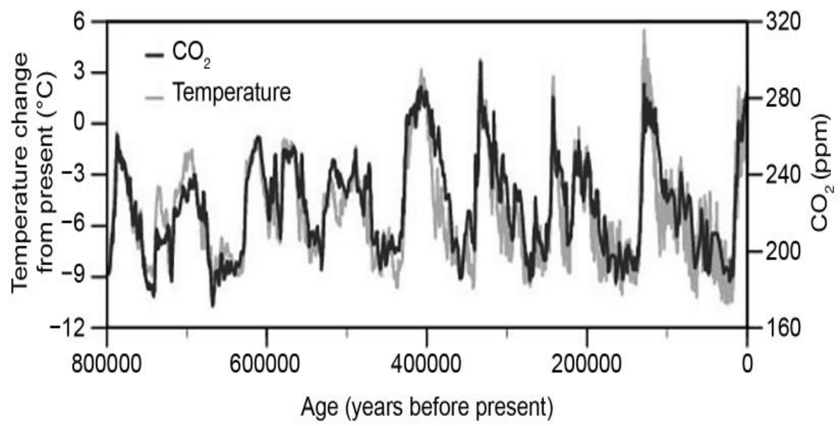
- Seasonal contrasts are more extreme in one hemisphere
  - Currently: S. Hem sees more extremes, while N. hem is more moderate
  - In 13,000 yrs: N. Hem will see more extremes
- Caused by tidal forces
- North stars will eventually change

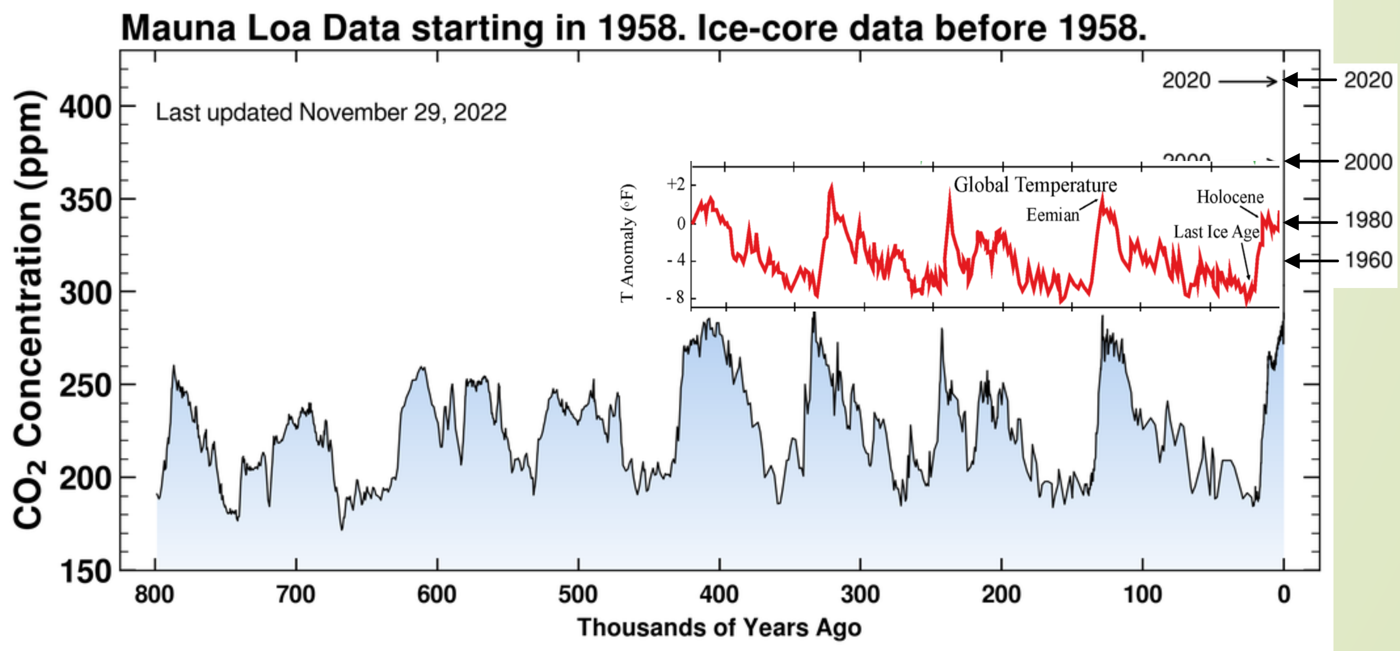


Summary: Orbital Cycles and Climate

- Orbital cycles = periods of glaciation and deglaciation
  - Cycles of T and CO_2
- But long timescales (23-100 kyr)= little difference in last century
- → Cannot explain rapid rise in CO_2 and temp in last century







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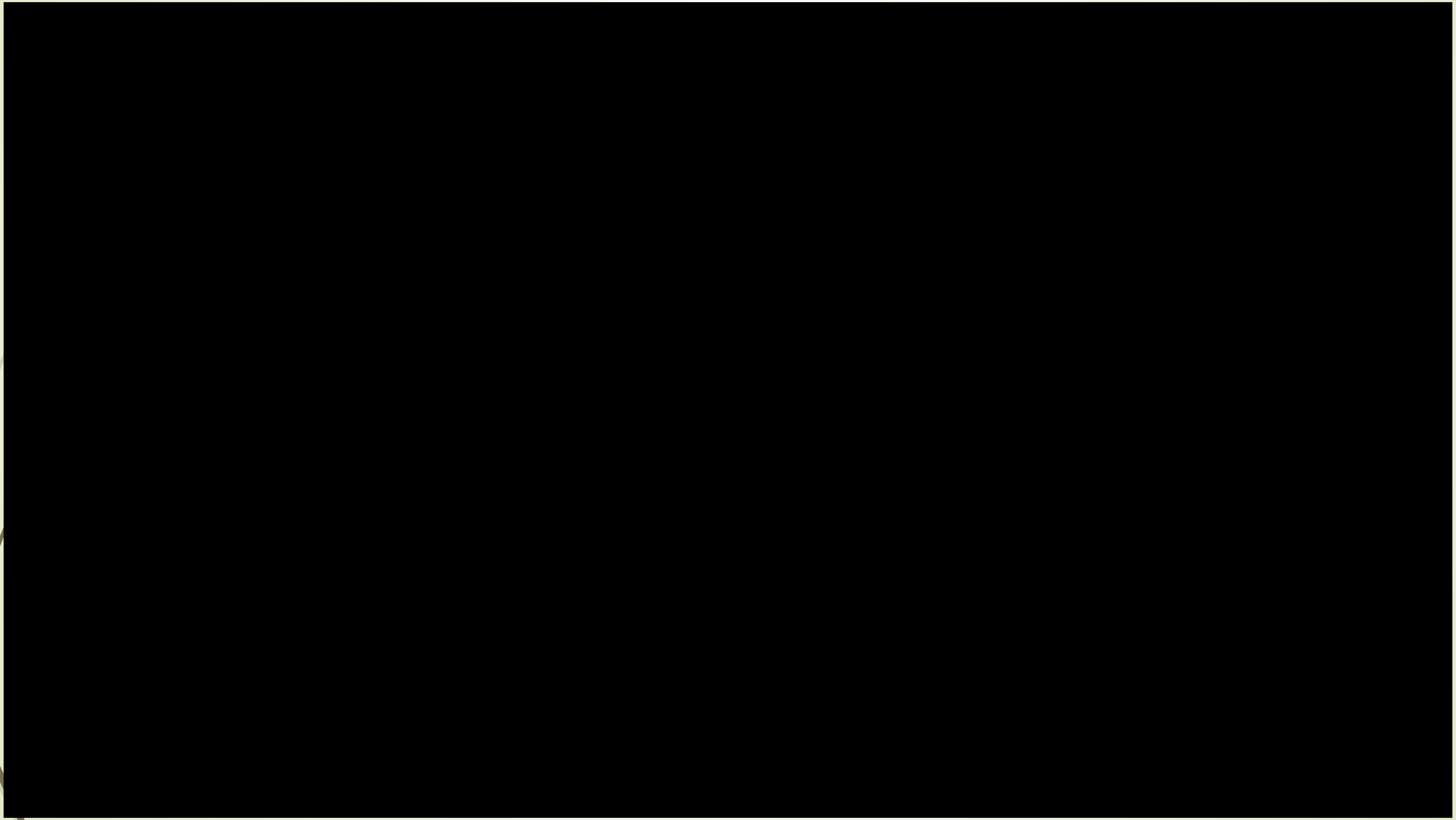
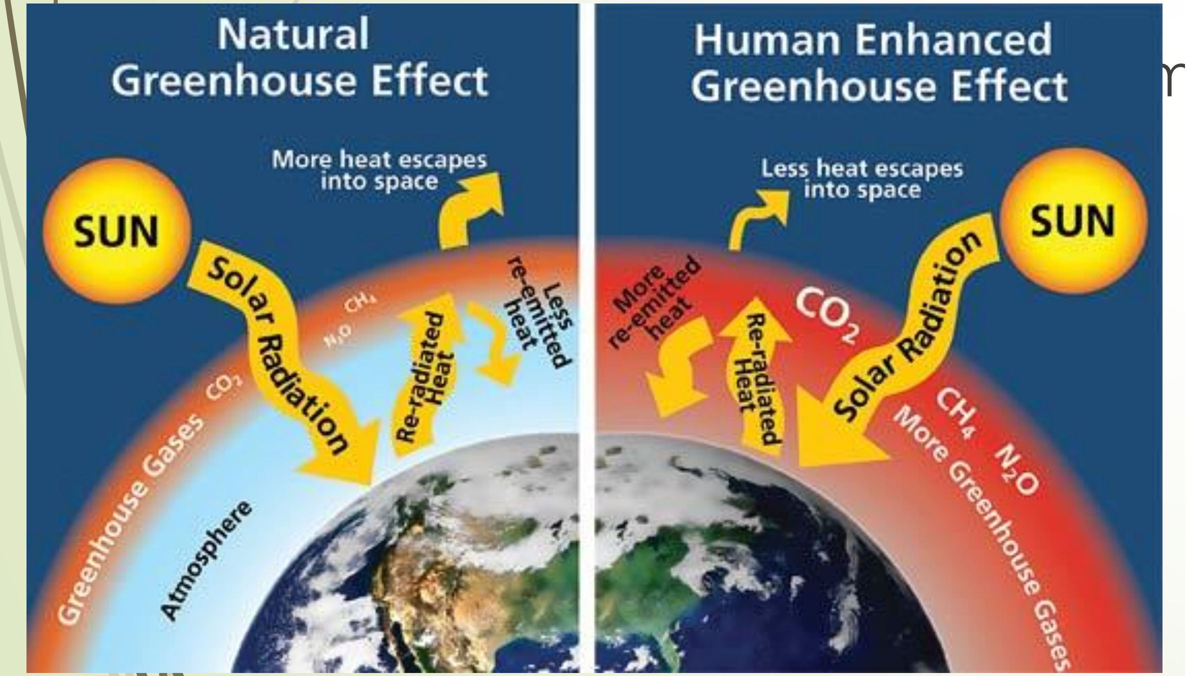
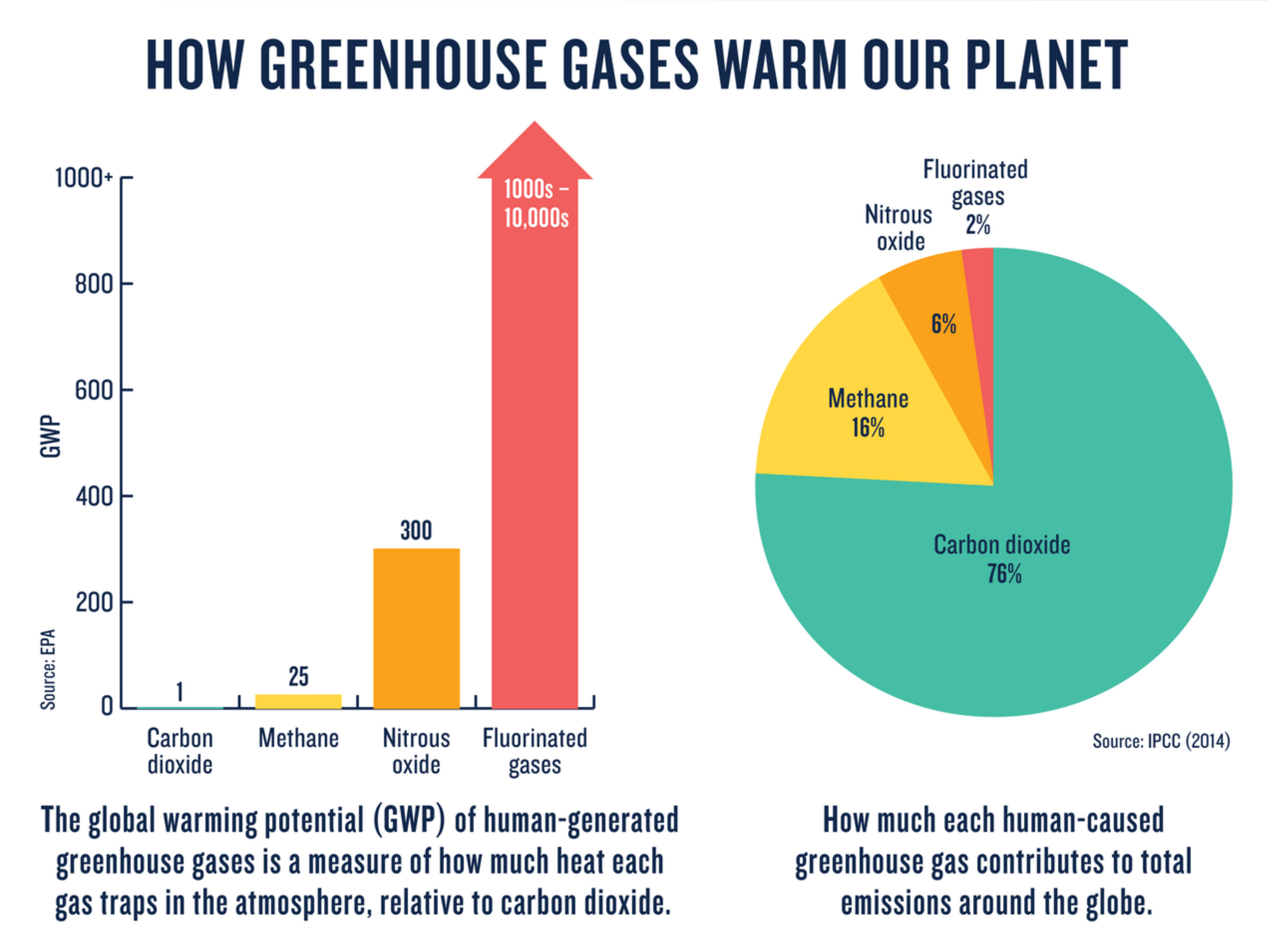
LO5: Earth’s Energy Balance



LO5: Earth's Energy Balance, Greenhouse Gasses

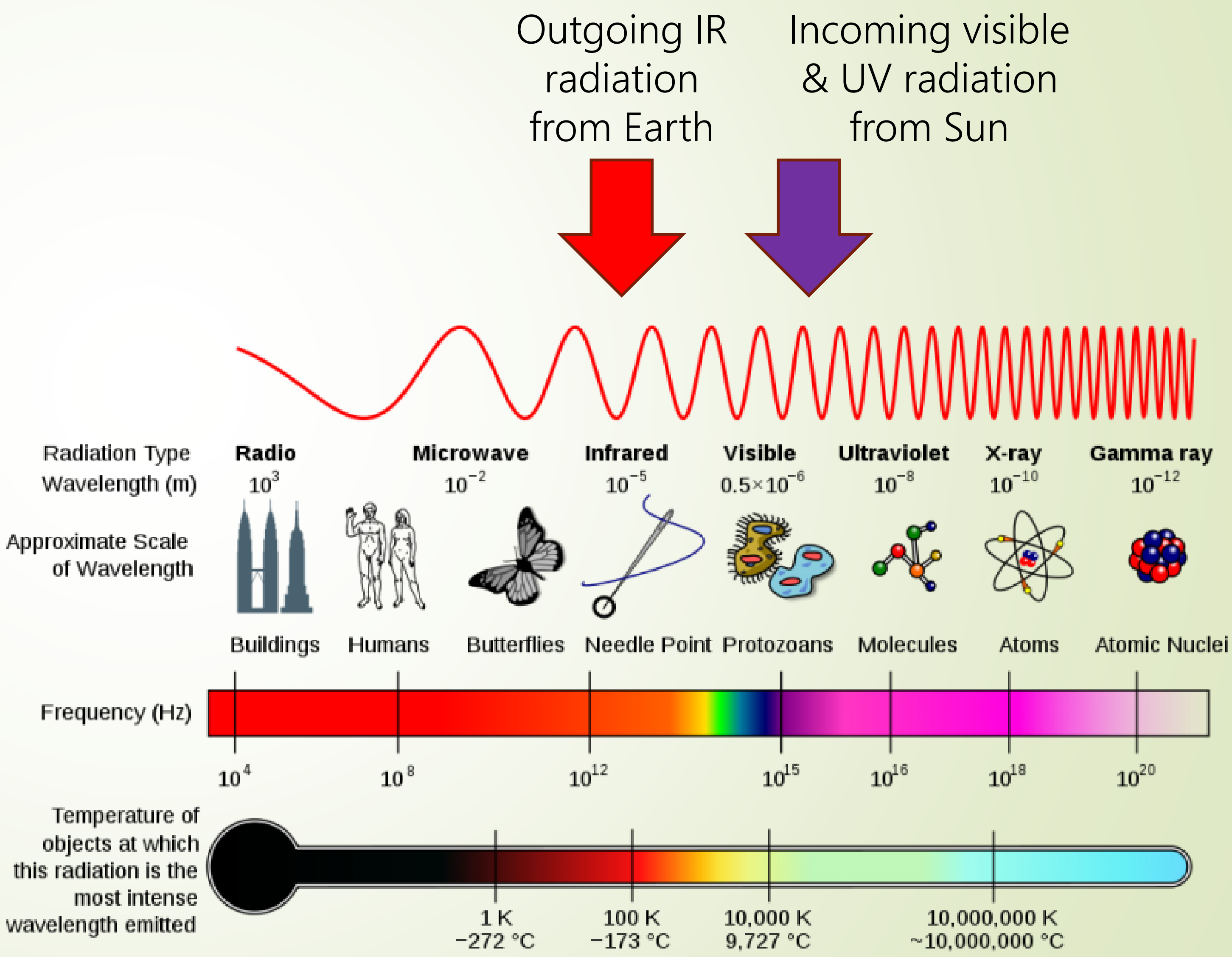
- Other GHG (e.g. N_2O, CO2





Quick background: Electromagnetic spectrum

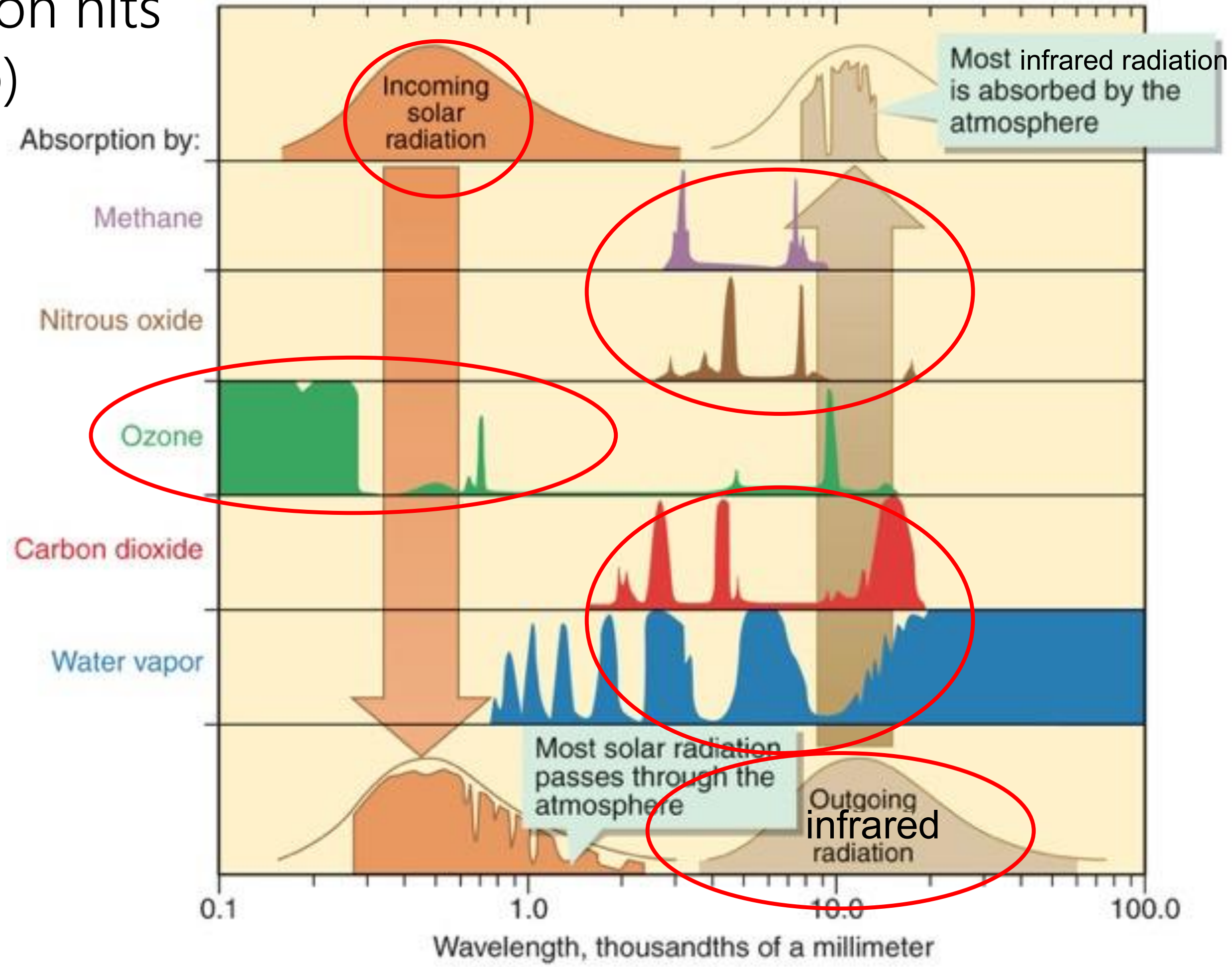
- Electromagnetic energy travels in waves, with different wavelengths
- *Thermal* radiation has different wavelengths depending on temp.
  - Thermal radiation of animals is in the IR spectrum (why we call IR radiation "heat")
- Solar radiation is UV (short wavelength)
- Earth absorbs and then re-emits this energy as "heat" in the IR spectrum (long wavelength)



LO5: How do greenhouse gases trap heat?

Incoming "short-wave" (UV) solar radiation hits Earth (heats it up)

Some is absorbed by oxygen (O_2) and ozone (O_3)



"shortwave" "longwave"

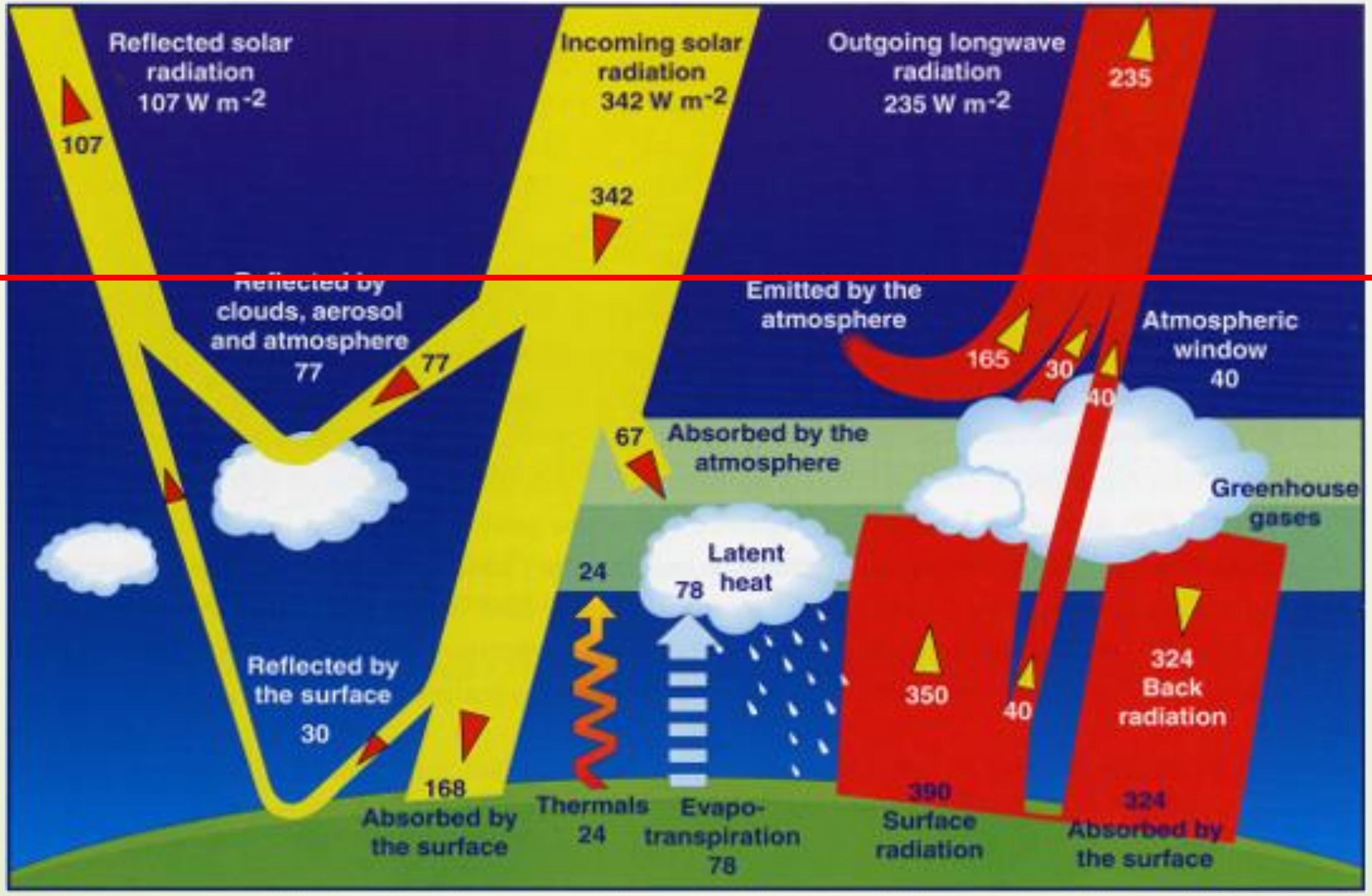
GHG: CO_2, Methane (CH_4), Nitrous oxide (NO_2), H2O absorb "long-wave" radiation → traps heat and "back-radiates"

Some of this radiation is re-emitted by Earth as long-wave (IR) radiation (heat)

LO5: Earth's Energy Balance

Earth's energy balance: The balance between incoming and outgoing solar radiation

Energy In (Shortwave) = Energy Out (Longwave)



LO5: Earth's Energy Balance

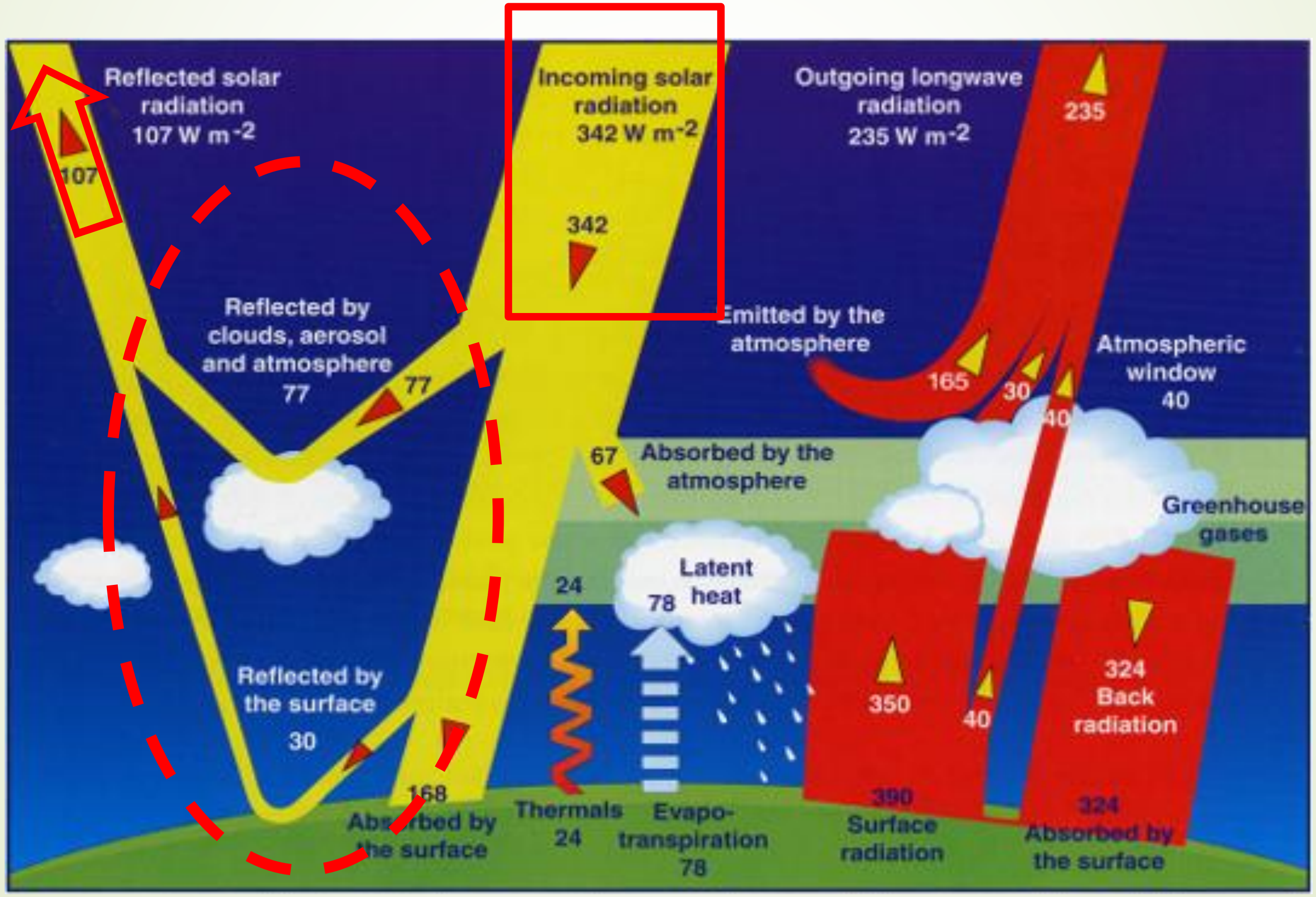
When energy comes in, where does it go?

Albedo

Net effect of the ***reflectivity*** *of Earth - makes the Earth* ***an imperfect absorber***

Reflectivity of Earth: 31% of light is reflected back to space

Albedo examples:
High: ice, snow, clouds
Low: forests, pavements



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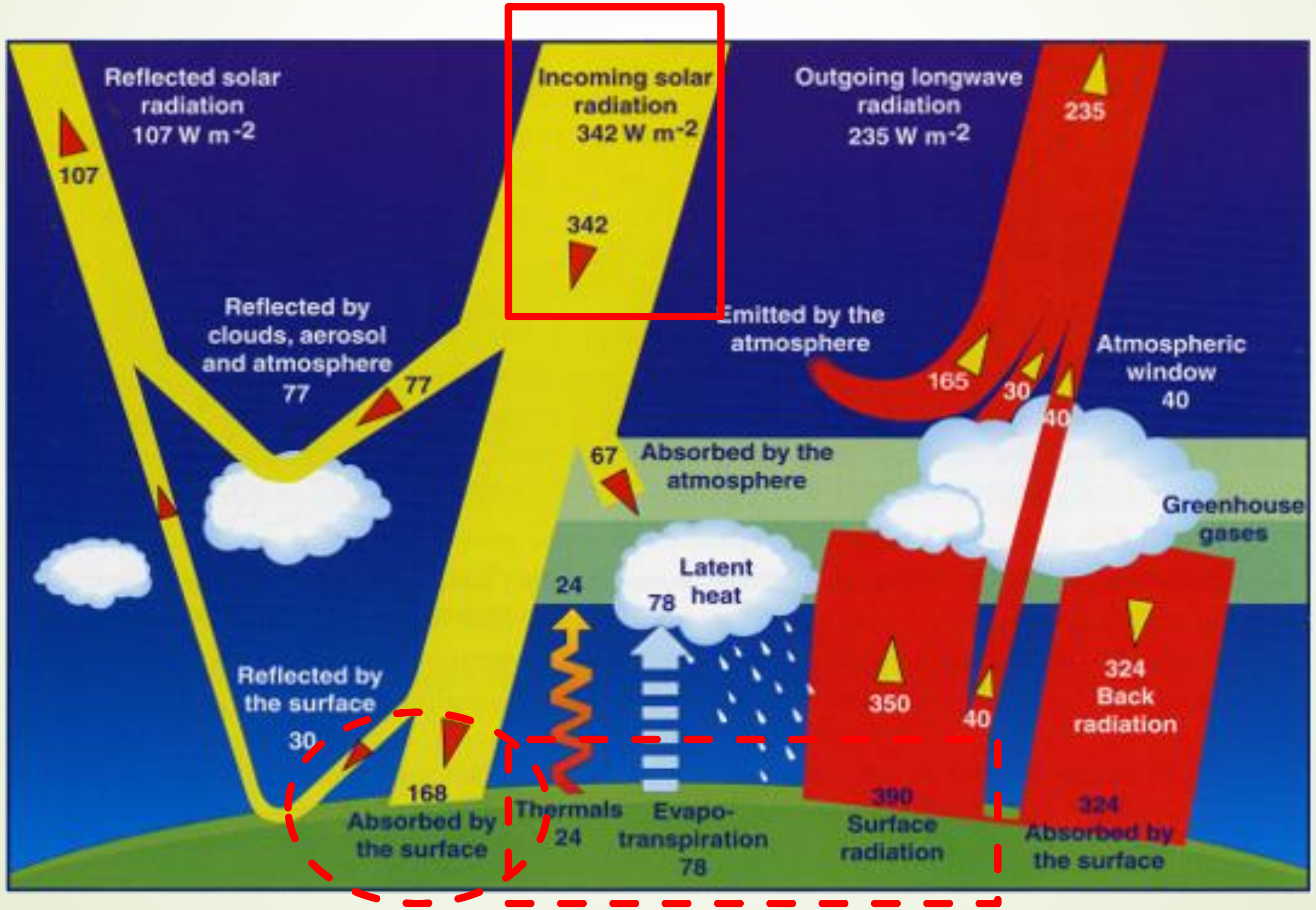
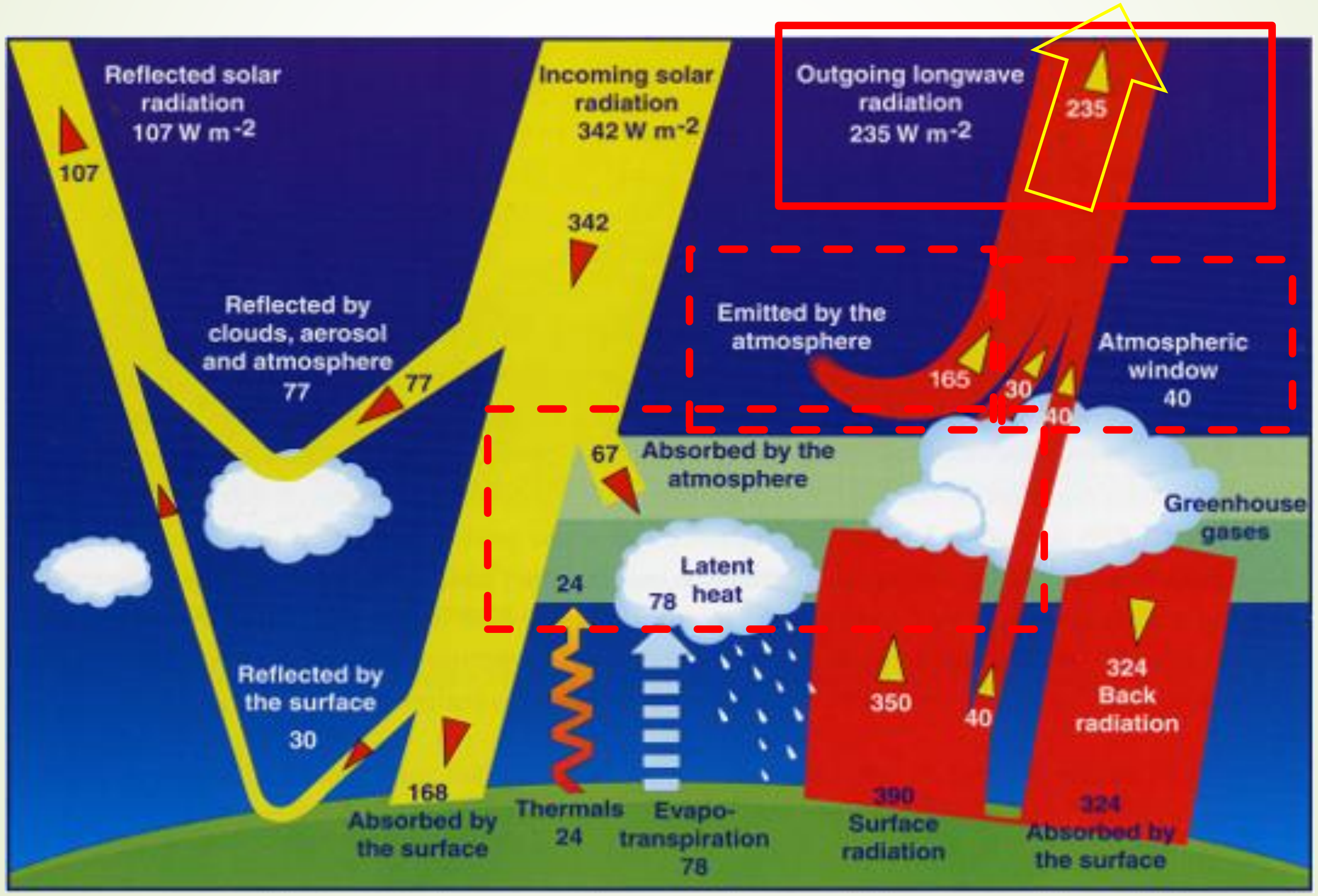


LO5: Earth's Energy Balance

When energy is within the atmosphere, where does it go?



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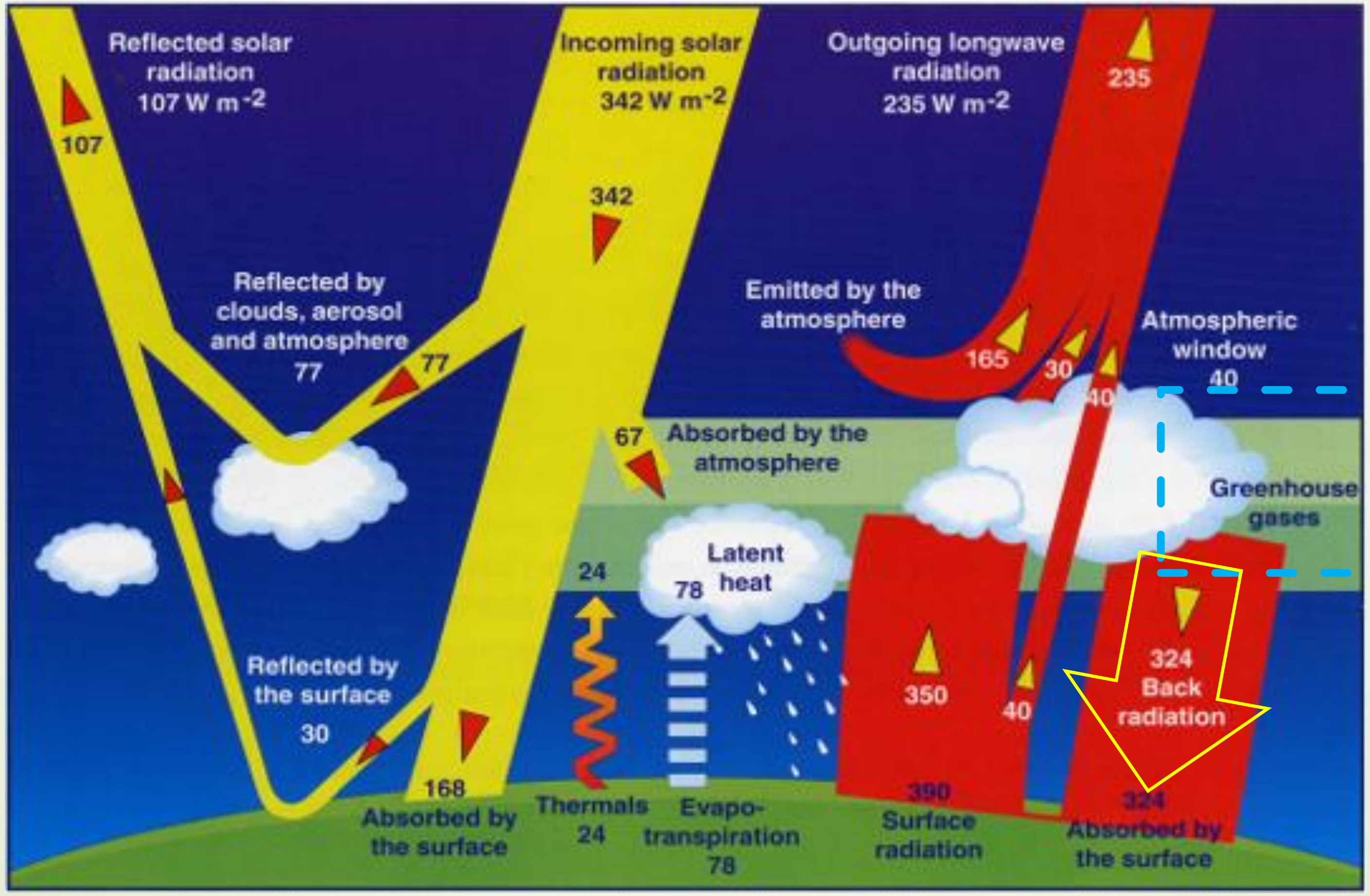
When energy is within the atmosphere, where does it go?



Greenhouse gases (GHGs)

Make the Earth an ***imperfect emitter***

GHG effect:

Process by ***which Earth warms*** *due to the composition of gases in the atmosphere*

GHG examples: CO_2 (carbon dioxide), CH_4 (methane), and H_2O_v (water vapor)

LO5: Earth's Energy Balance

- **Climate (radiative) forcing**: a factor that influences Earth's energy balance (natural or anthropogenic)

Albedo

*Net effect of the **reflectivity** of Earth - makes the Earth **an imperfect absorber***

Reflectivity of Earth: 31% of light is reflected back to space

Albedo examples:
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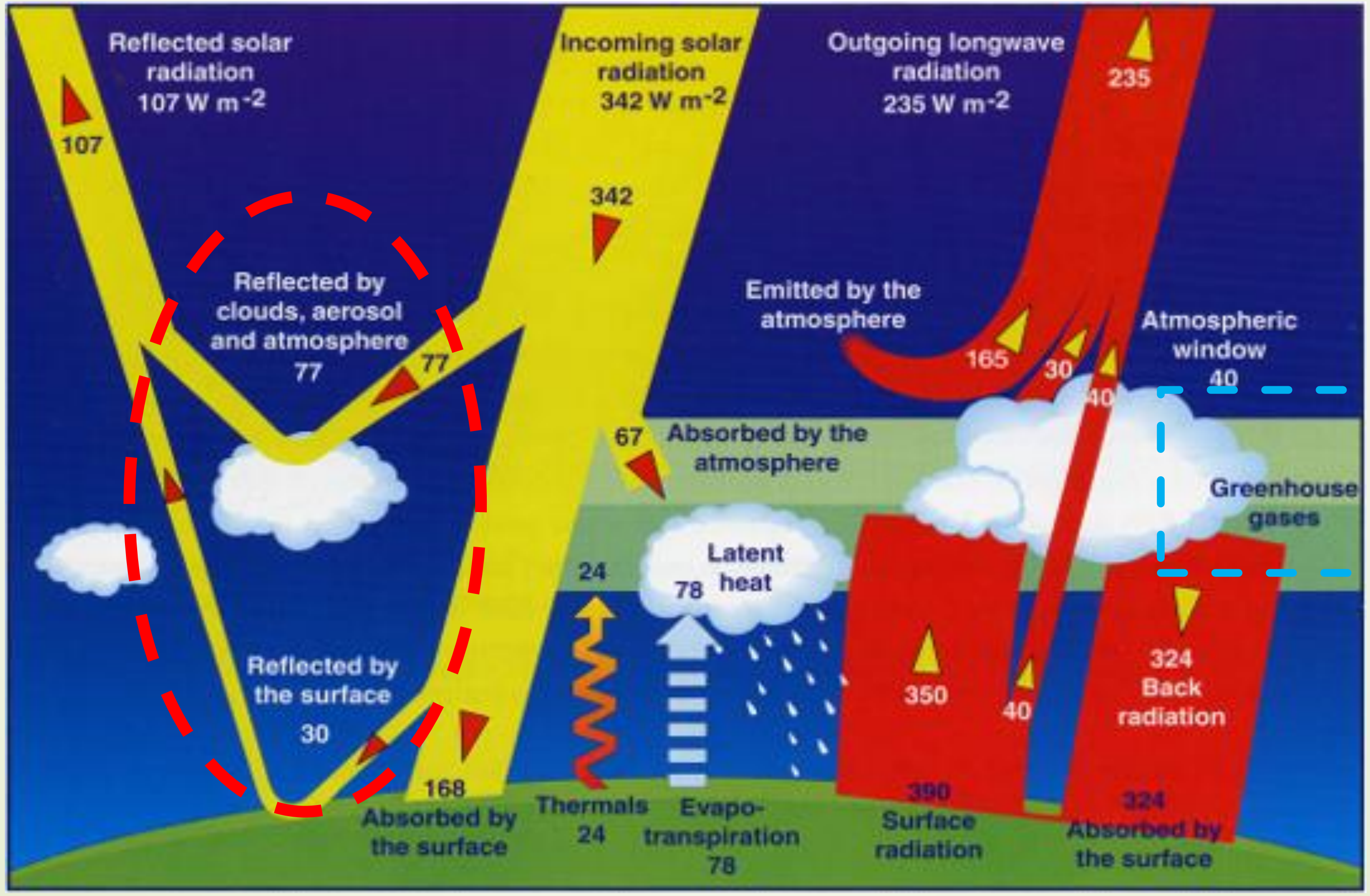

Greenhouse gases (GHGs)

*Make the Earth an **imperfect emitter***

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*Process by **which Earth warms** due to the composition of gases in the atmosphere*

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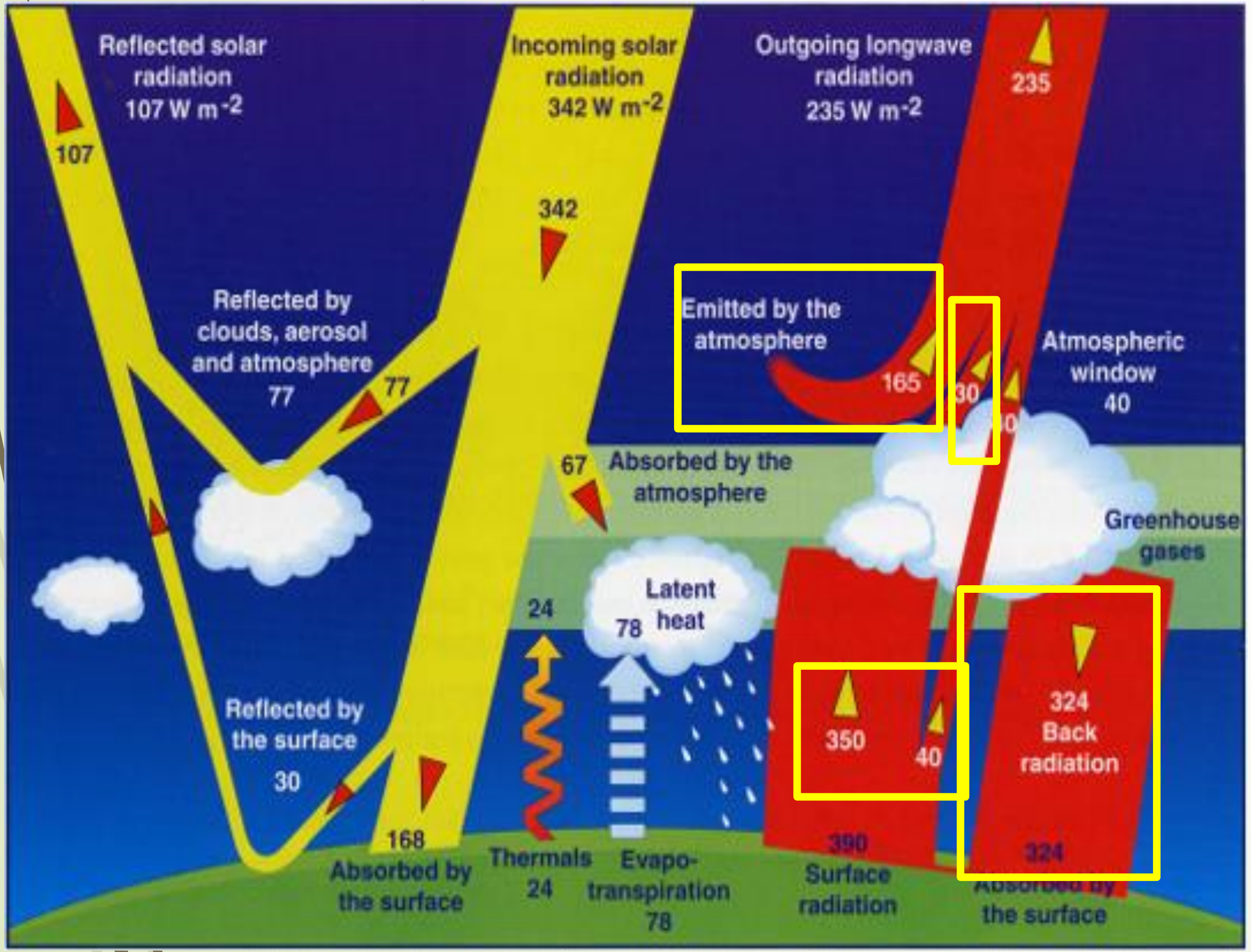
LO5: Energy Balance - Questions



WARM-UP

Energy emitted by:
atmosphere? 165 W/m^2
clouds? 30 W/m^2

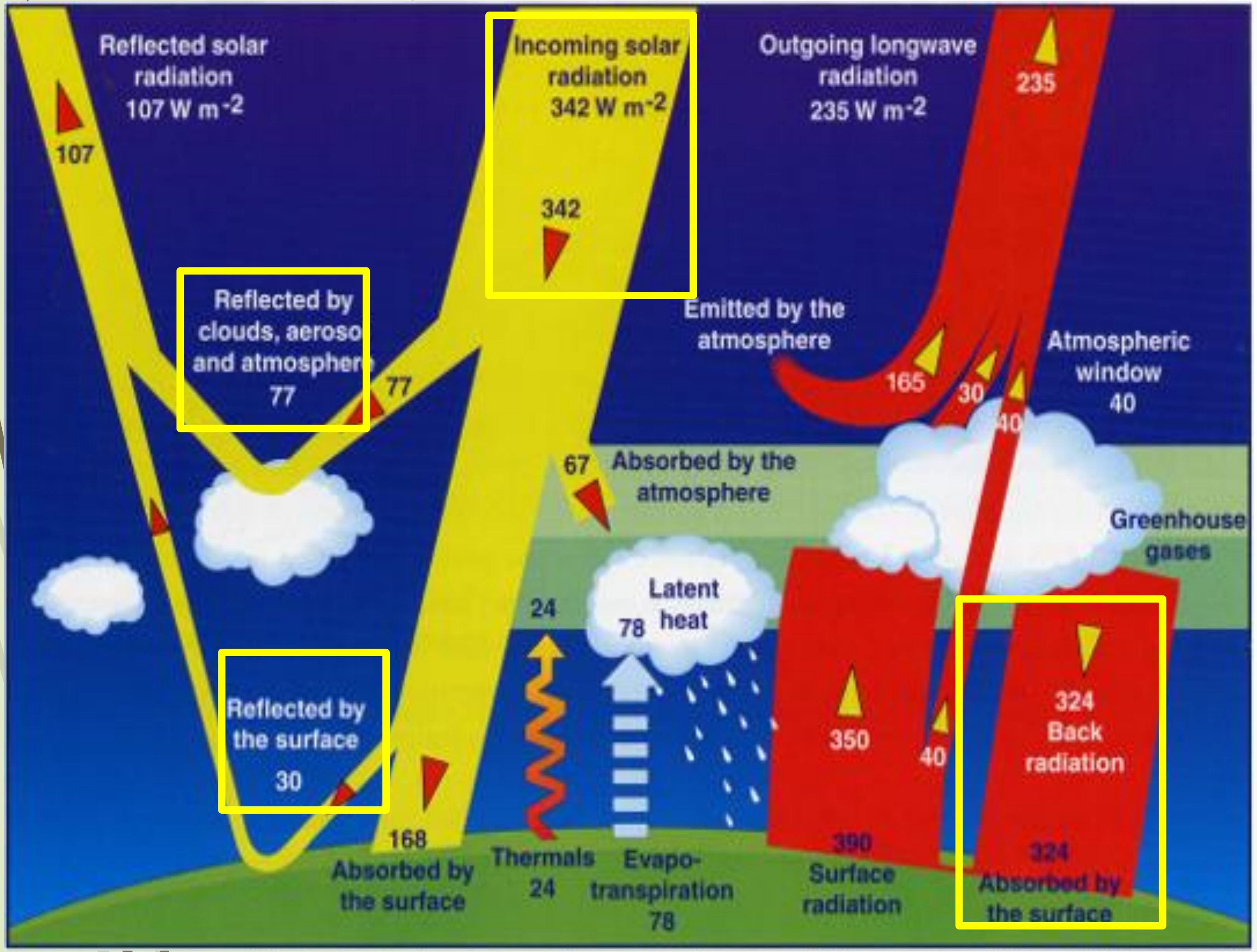
What is the *surface radiation* (energy emitted by the surface)?

390 W/m^2

How can 390 W/m^2 be *emitted by the surface* when only 168 W/m^2 is *absorbed by the surface* from the Sun?

Because there's also 324 W/m^2 in back radiation!
168+324=492 W/m^2

LO5: Energy Balance



WARM-UP (cont.)

Calculate how much solar energy is reflected back:
30+77 =107 W/m2

List different sources of energy absorbed by Earth:
- Incoming solar radiation
- Back radiation from the atmosphere

Which of them has the greatest radiation absorbed by Earth?
Back radiation from the atmosphere

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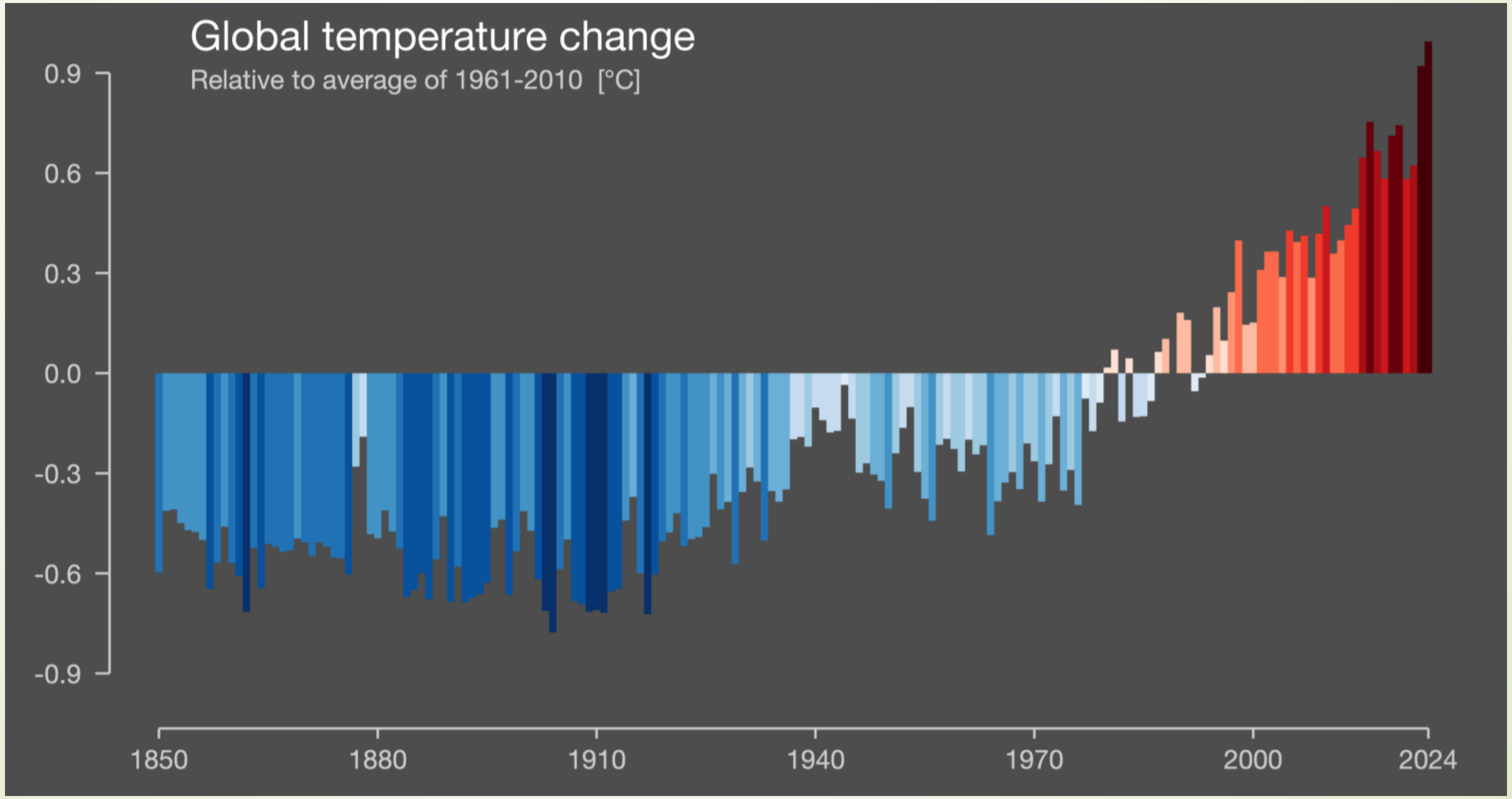
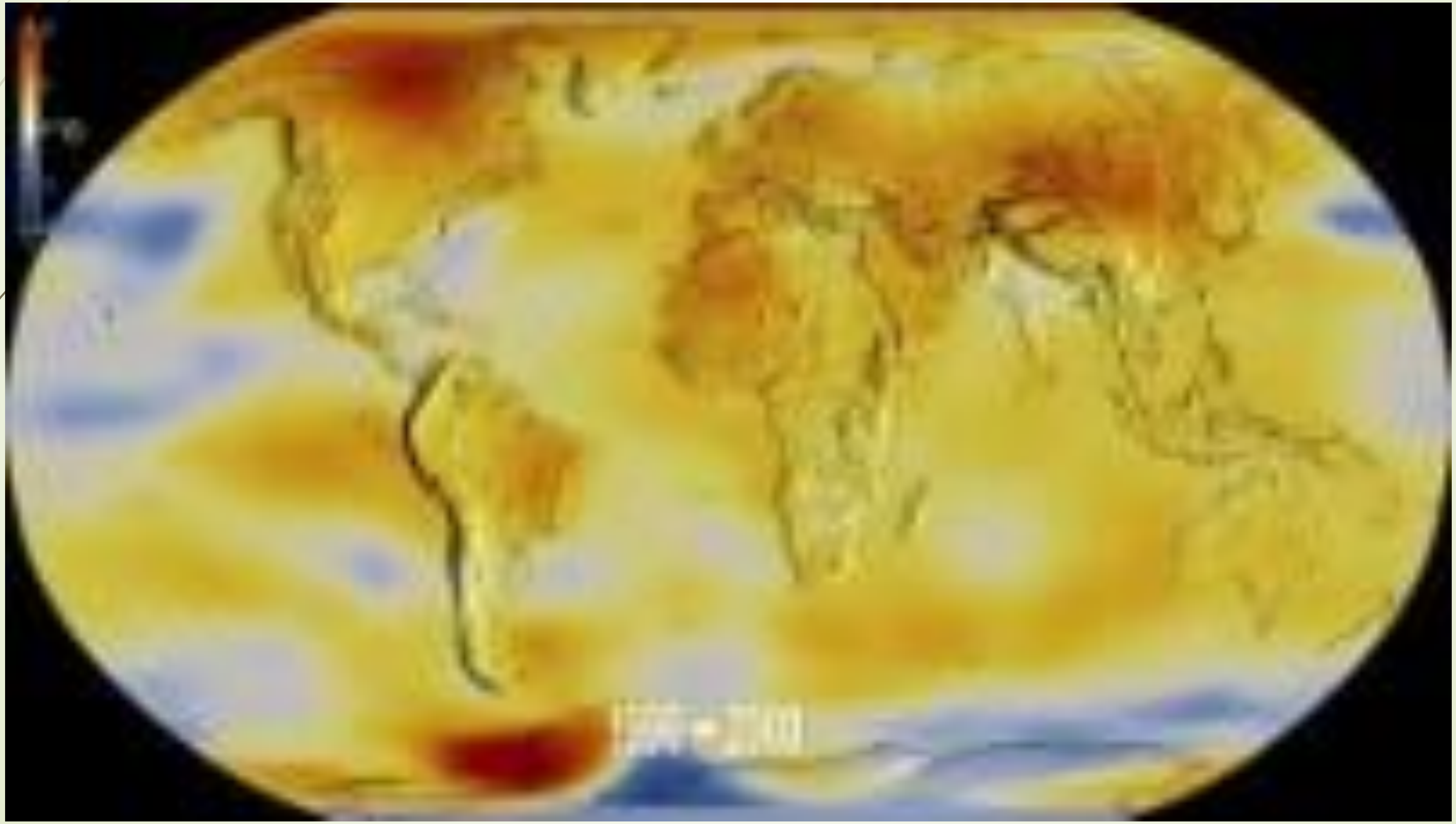
LO4: Global Temperature record since 1880

- The combined land and ocean temperature has increased at an average rate of **0.07°C (0.13°F) per decade since 1880**
- The average rate of **increase since 1981 is more than twice that rate**: 0.18°C, (0.32°F)



LO4: Global Temperature record since1880



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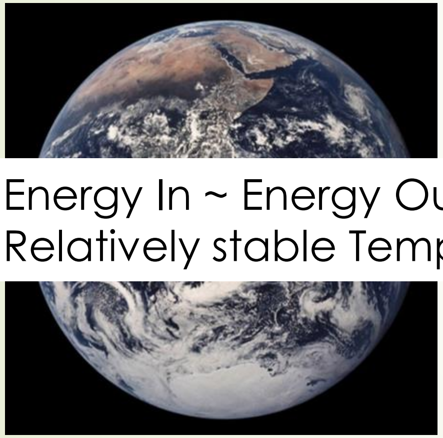
LO3: Atmospheric composition

Atmospheric composition plays major role in energy budget

Earth's Atmosphere:

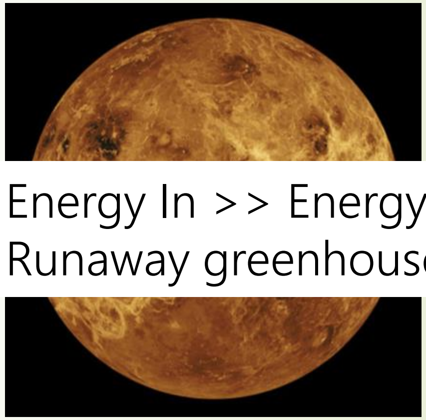


- Energy In ~ Energy Out
- Relatively stable Temp

Similar levels of solar radiation

Different energy budgets

Venus' Atmosphere:



- Energy In >> Energy Out
- Runaway greenhouse

Earth:

- Nitrogen (N_2) 78.1%
- Oxygen (O_2) 21.0%
- Argon (Ag) 0.9%
- Water vapor (H_2O) ~ 2.5%
- **Carbon dioxide (CO_2) 0.04%**

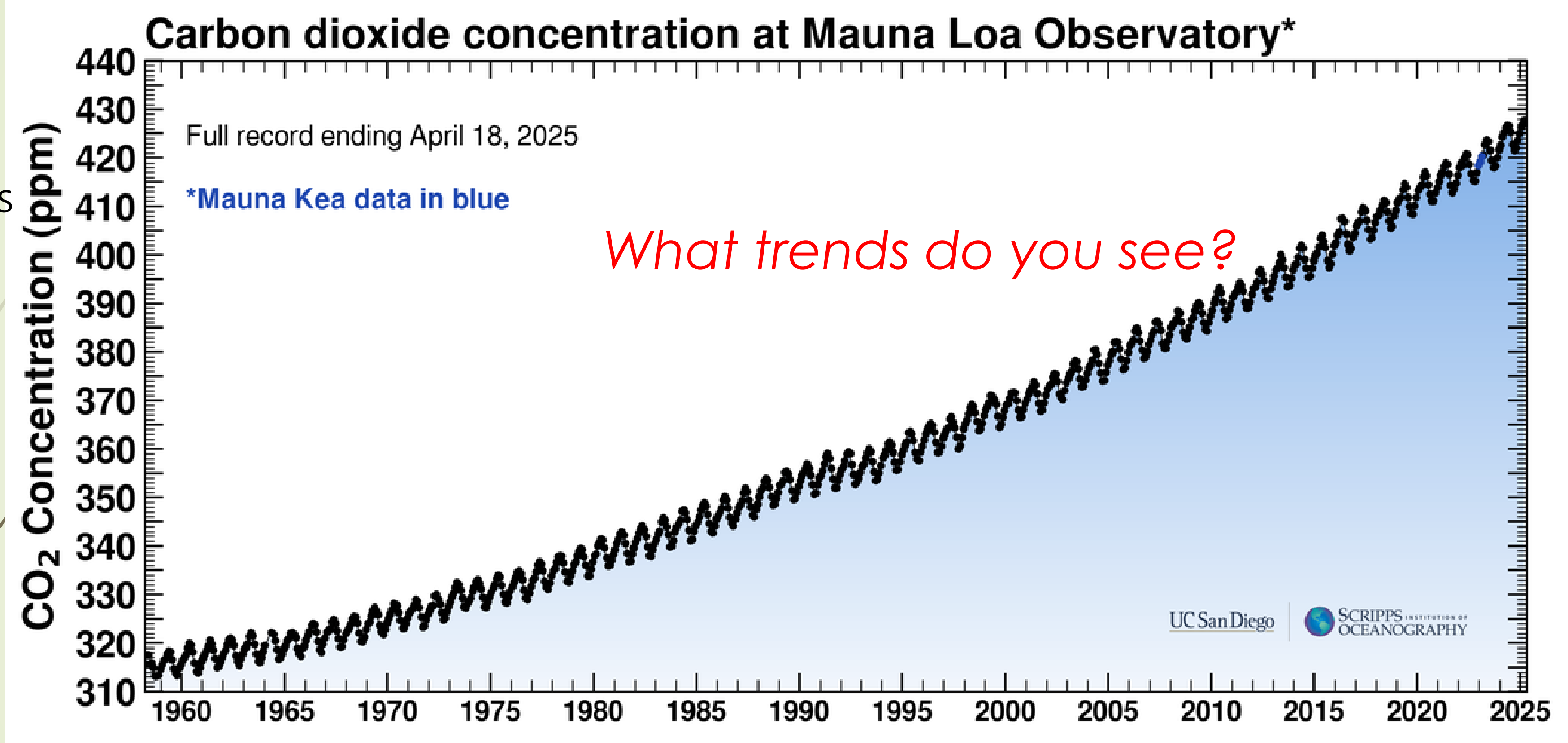
Venus:

- **Carbon dioxide (CO_2) 96.5%**
- Nitrogen (N_2) 3.5%

What recent changes do we observe?

11/12/25: 426.77 ppm

Earth's atmospheric CO2 content is increasing at **0.6% per year** (Also rises in N_2O and CH_4)

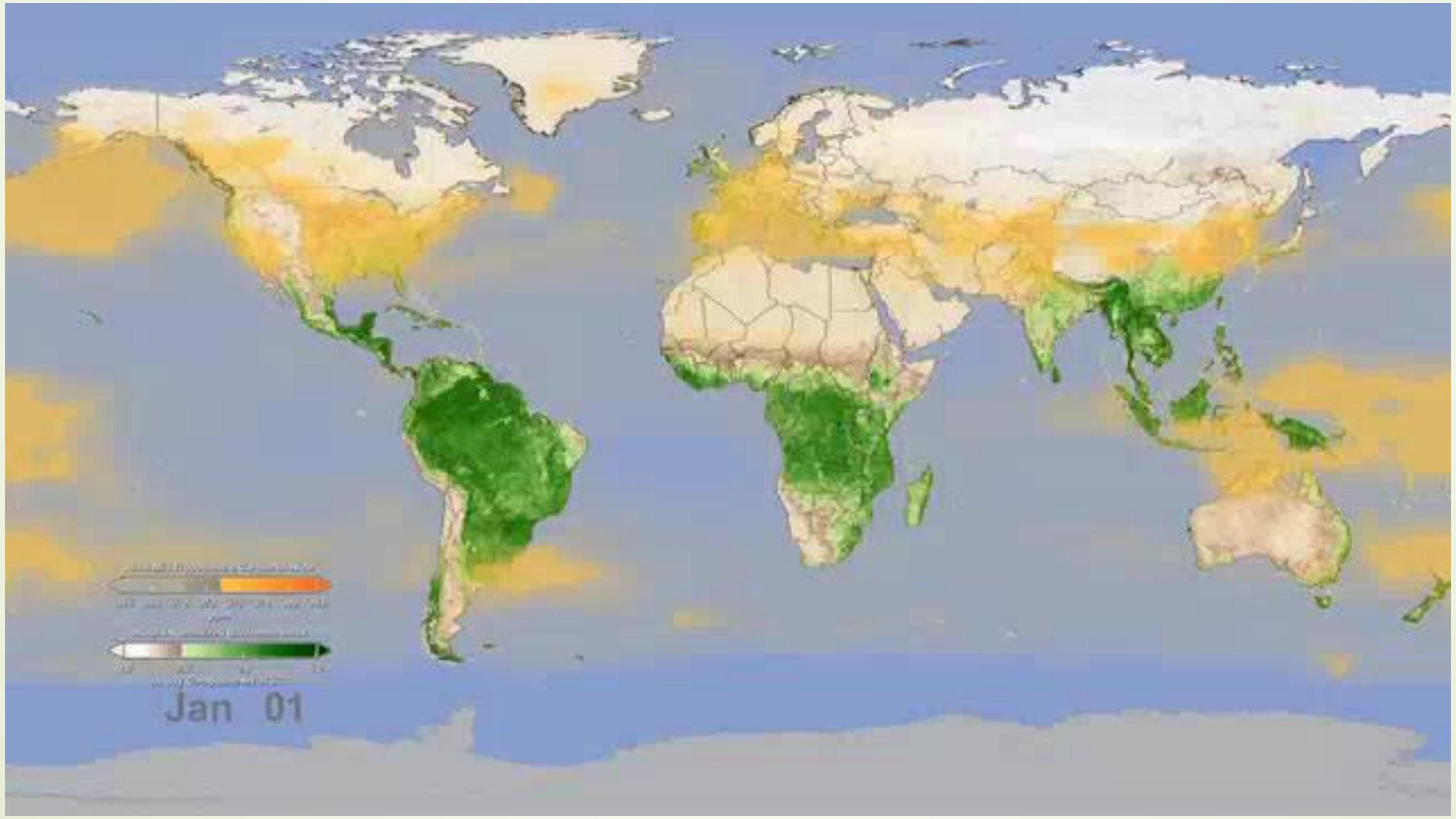


Intra-annual changes in atmospheric CO2 due to the massive "greening" of the N. Hem. during the months of May – October. = ↓CO_2

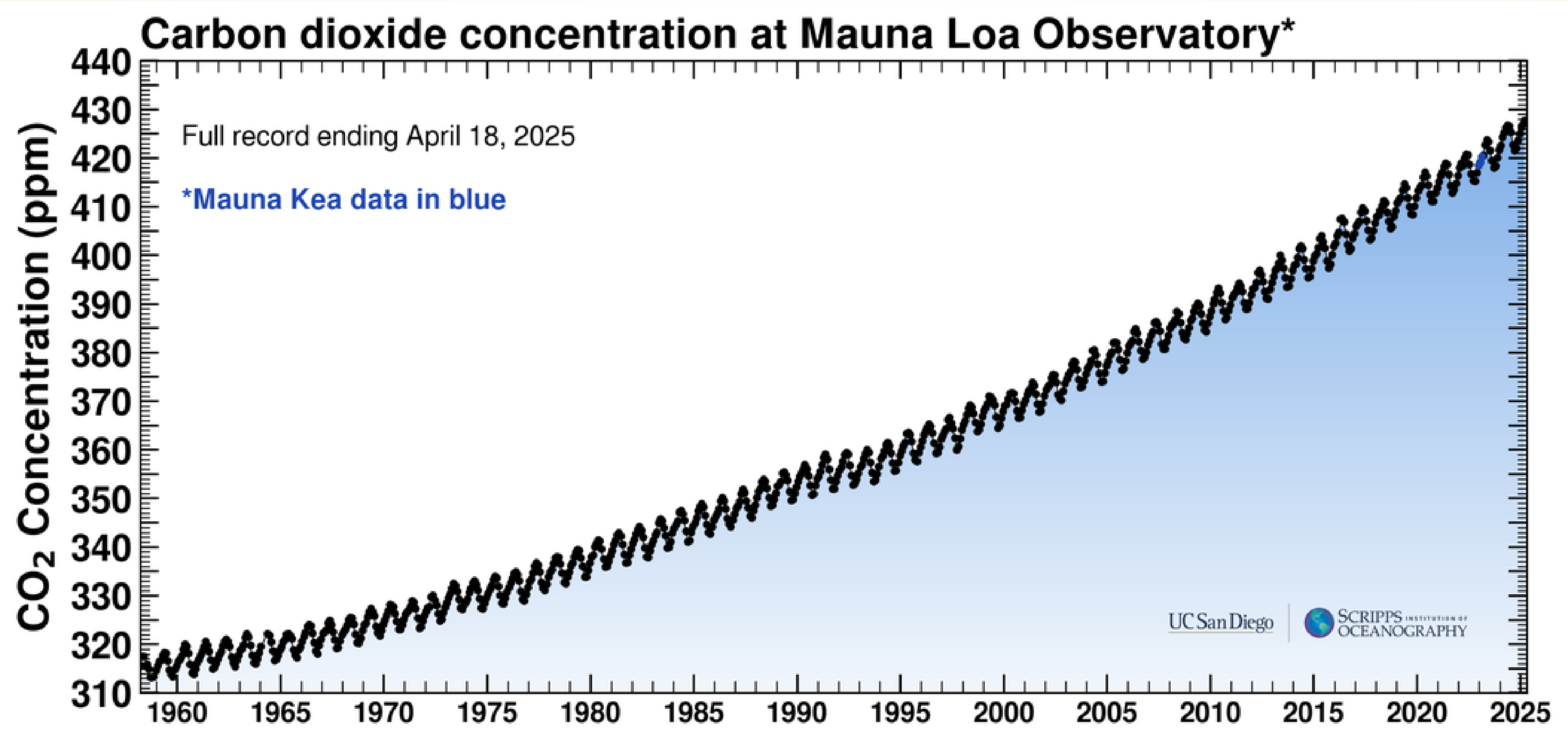
During all of the cycles between ice ages and warm periods over the past million years, atmospheric carbon dioxide **never climbed higher than 300 parts per million**.

Intra-annual changes in CO2

Intra-annual changes in atmospheric CO_2 due to the massive "greening" of the northern hemisphere during the months of May – October.



CO2 concentration in the atmosphere is INCREASING

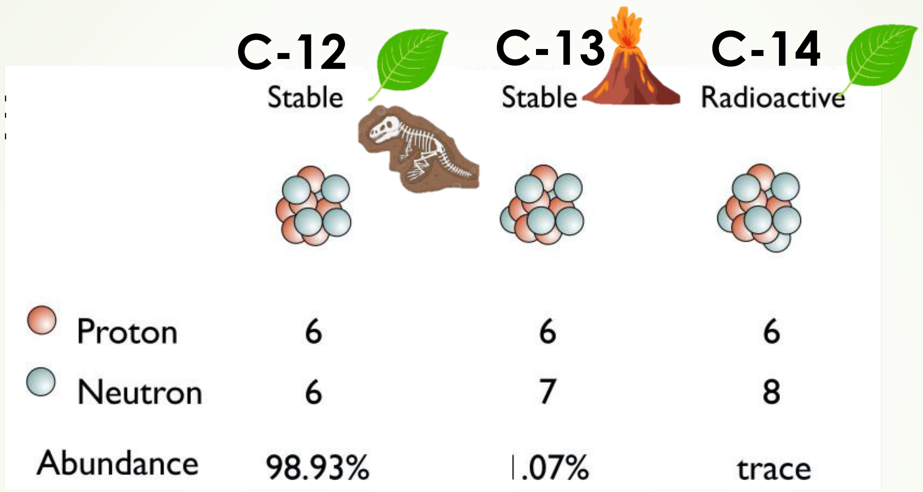


WHAT ARE THE SOURCES OF THIS CO2?

Isotopes

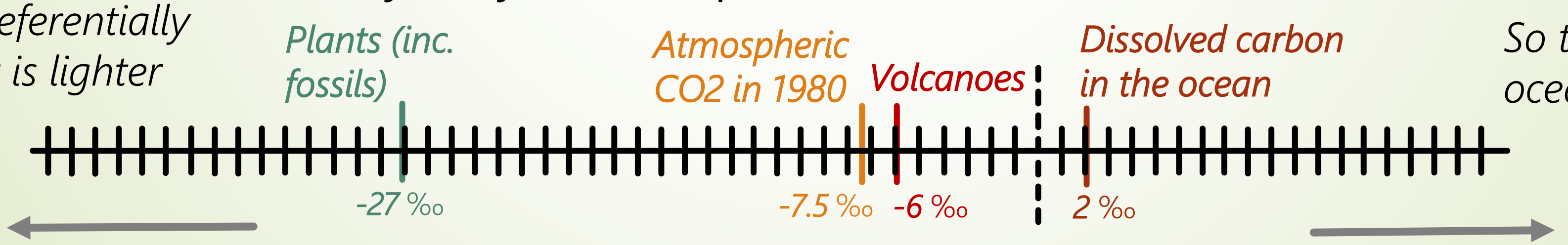
Isotope = Same element Differing numbers of neutrons

The isotopic composition of the carbon in carbon dioxide helps us identify the source of the carbon



$\delta^{13}C$ notation: used to relate carbon isotope values (C-12 and C-13) on a standardized scale (easy way to compare different sources of CO2)

Plants (& plankton) preferentially take up 12C because it is lighter



So there is more 13C leftover in ocean and atmosphere

Lower $^{13}C/^{12}C$ ratio

$\delta^{13}C$ of CO2 (relative to a standard, given in unit of ‰)

Higher $^{13}C/^{12}C$ ratio

FYI: Why do volcanoes have higher ^{13}C?

Why do volcanoes have heavier C (^{13}C-rich) compared to plants?

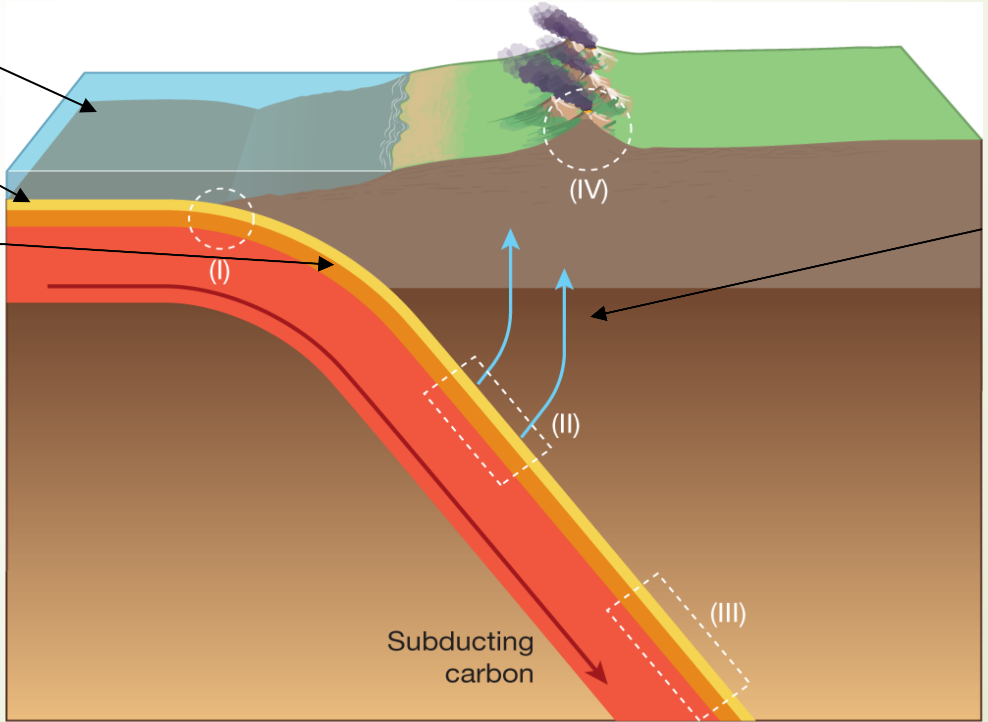
- Plate Tectonics bringing ^{13}C-rich sediment (made of shells) into the mantle, which melts → ^{13}C-rich CO2

(5) ^{13}C-rich CO_2 erupted at volcanoes

(1) Phytoplankton take up ^{12}C= Ocean is enriched in ^{13}C

(2) ^{13}C-rich shells sink and form limestone ($CaCO_3$)

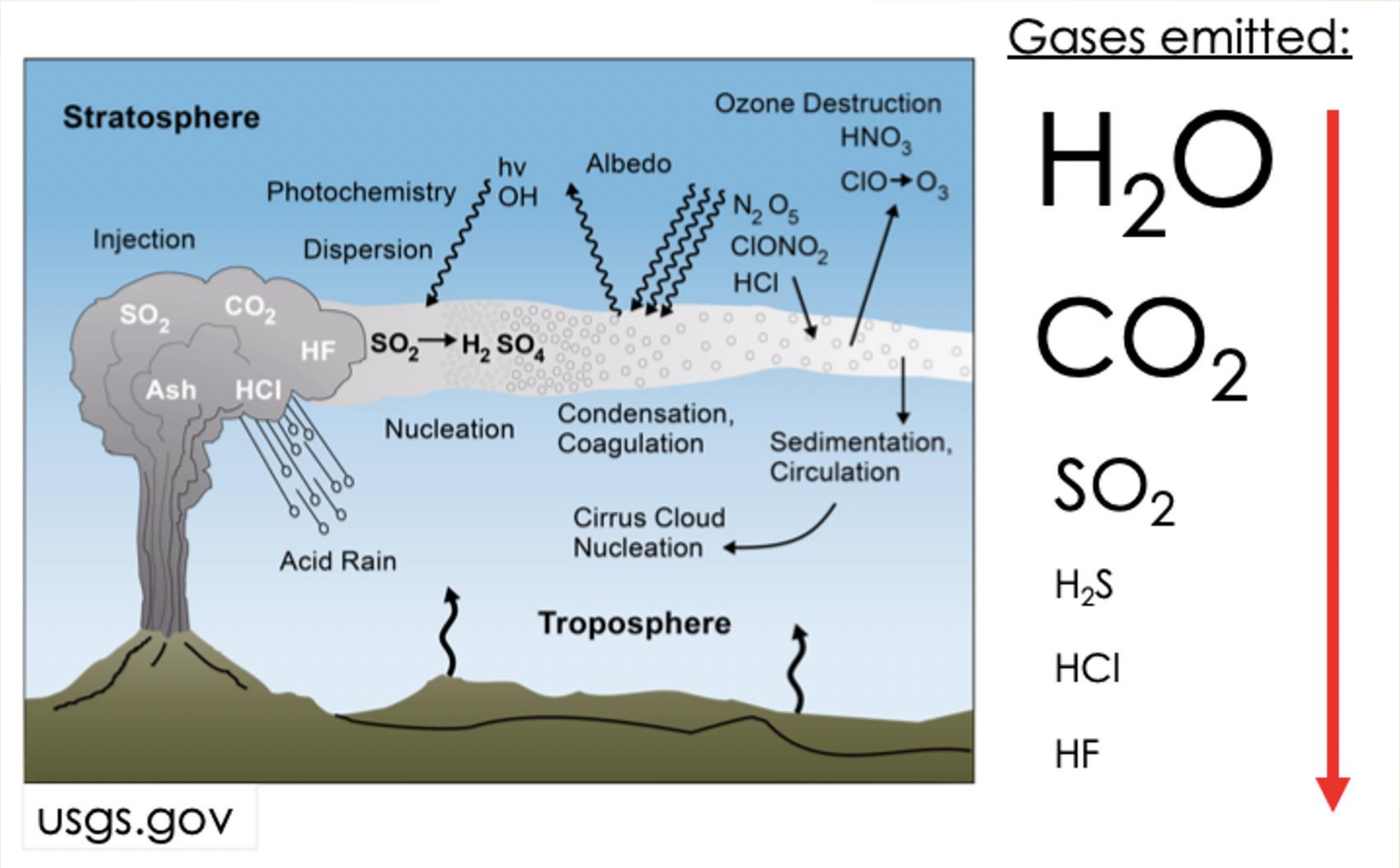
(3) Oceanic plate gets subducted into the mantle



(4) ^{13}C- rich limestone melts=^{13}C-rich CO2

Rising CO2: What is the cause?

Could volcanoes be the cause of recent global warming?



Activity: Collecting Climate Change Data

Option 1 – harder!

Getting Started

Form groups of 3

Read through the instruction handout

Journal Entries

Use the first set of clues to fill in the blanks of the journal entry (including any graphs and tables you encounter!)

When you have finished, wave me over to get your answers checked before receiving the next set of clues

Letter

When you have finished the entire journal entry, craft a letter in response to your friend explaining your findings

Activity: Collecting Climate Change Data

Option 2 – more straightforward!

<u>Getting Started</u>

Form groups of 3

Open the **Google sheet** provided in this week's module as well as the **Google doc** handout

Make a copy of **both**, rename it with your names, and be sure to share it with your group members and with me

<u>Part 1: Isotopes</u>

Once you are set up, complete the **first two tabs** (Isotopes Part 1 and Isotopes Part 2)

Part II: Ice Cores

Ice cores are like a time machine!

- Ice (from ice sheets or glaciers) traps bubbles of air when it freezes
- These air bubbles are then preserved as tiny samples of ancient air
- This allows us to measure the composition of gases in the air at the time the ice was formed.
- We can use this as a way to directly measure changes in atmospheric CO_2 concentration over thousands of years.

Part II: Ice Cores

Part 2: Ice Cores

Go to **Tab 3: Ice Core Data** in the Google sheet

Use the diagram of the ice core to enter the CO_2 concentration and age (in thousand years before present) into the outlined red cells for all the selected ice core air bubbles. The graph will automatically be drawn as you enter the data (the first data point is entered for you).

Answer the corresponding questions on your handout

Part III: Volcanoes

Part 3: Volcanoes

Go to **Tab 4: Volcanoes SO2** in the Google sheet

- Read the excerpt from the NASA Earth Observatory site about SO_2 emissions and the effect on global and regional temperatures
- Answer the corresponding questions on your handout

Go to **Tab 5: Volcanoes CO2** in the Google Sheet

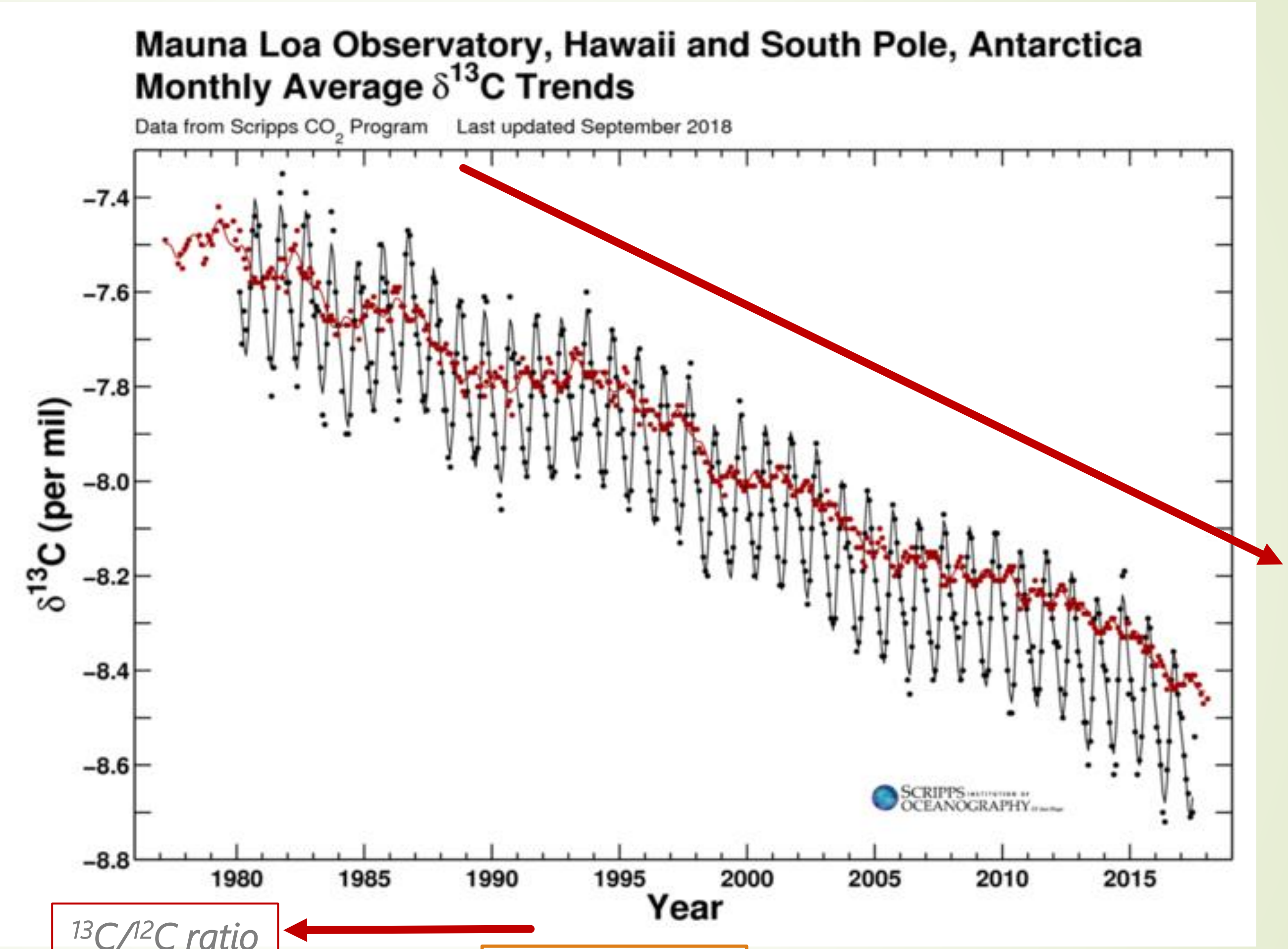
- Use the data provided to answer the corresponding questions on your handout

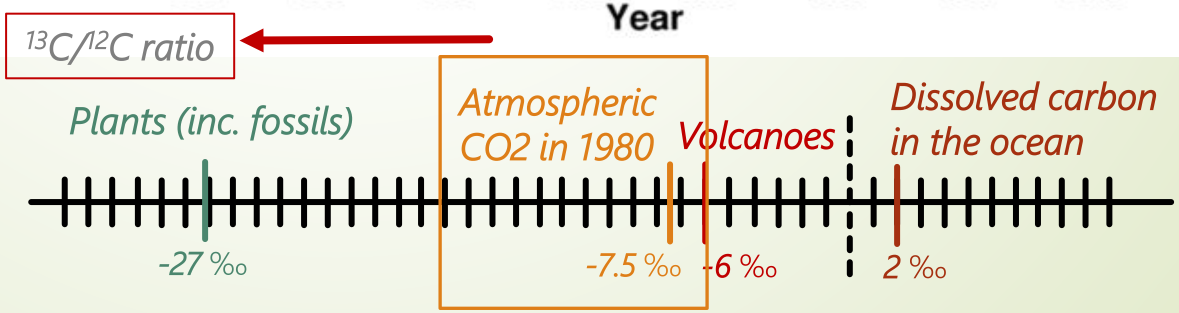
The $^{13}C/^{12}C$ ratio is decreasing

Atmospheric carbon tends to be "heavy" (lots of the heavier ^{13}C isotopes) relative to other carbon reservoirs, such as vegetation and fossil fuels.

The CO2 in the atmosphere is becoming richer in ^{12}C isotope relative to the ^{13}C isotope.

The $^{13}C/^{12}C$ ratio is decreasing, and the $\delta^{13}C$ is decreasing (becoming more negative)



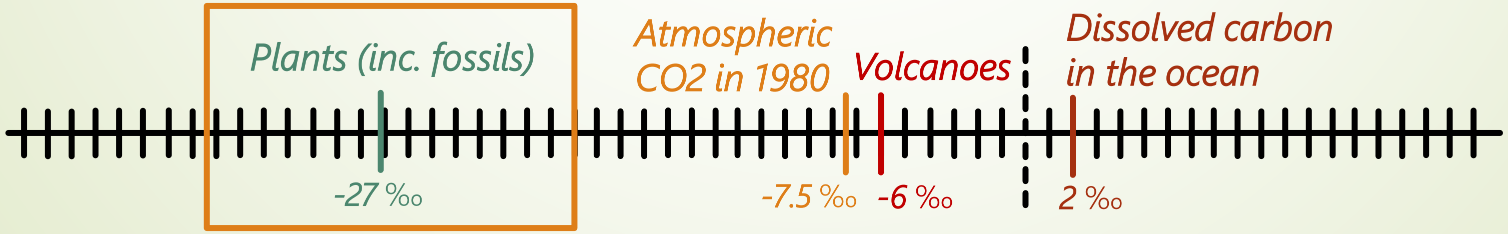


Where is the CO2 coming from?

So we must have a lot of new CO_2 being added to the atmosphere that has a **low $^{13}C/^{12}C$ ratio**

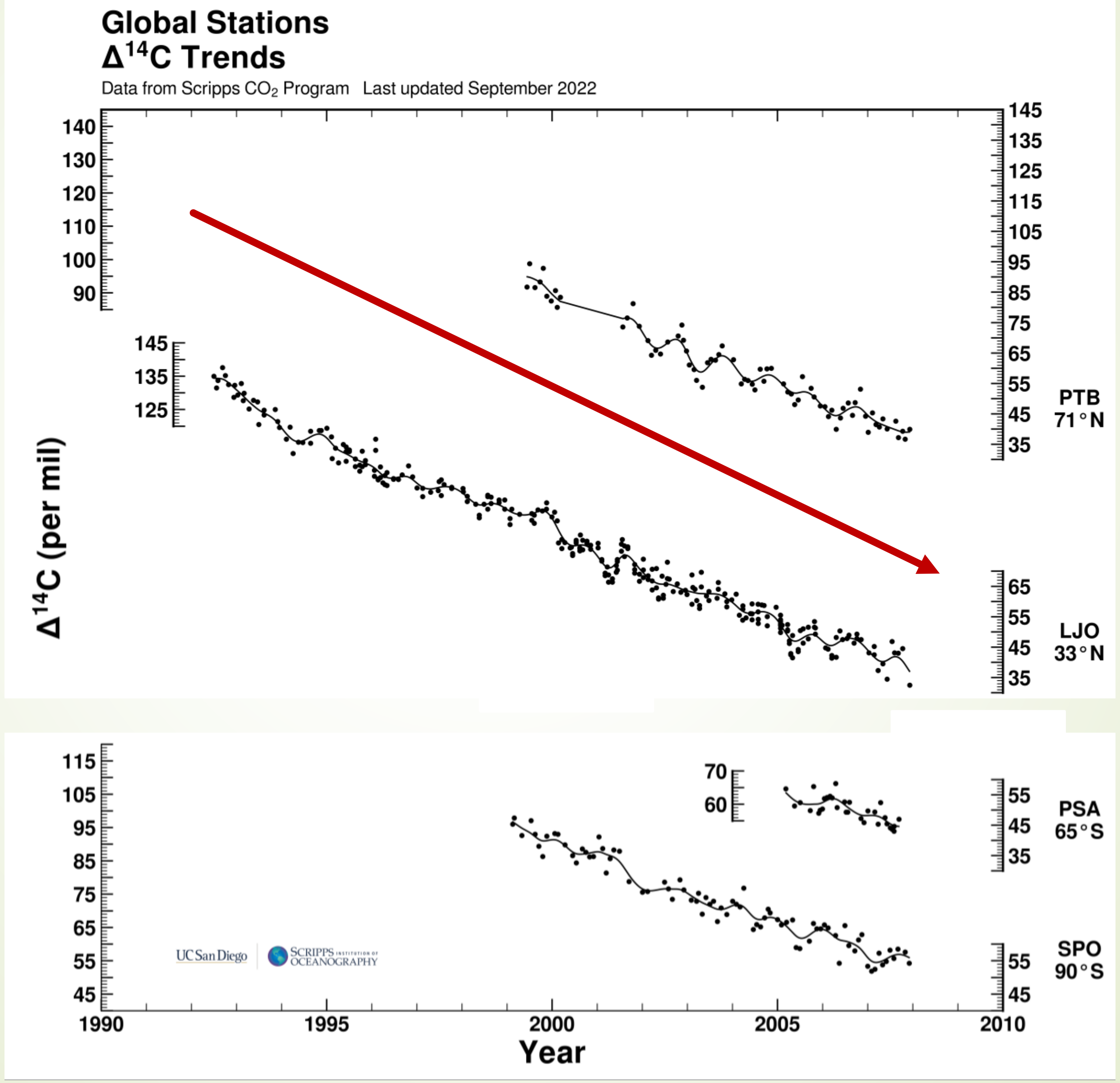
What has this ratio?

But: recent or ancient plants?



$\delta^{13}C$ of CO_2 (relative to a standard, given in unit of ‰)

FACT 3: the $^{14}C/^{12}C$ ratio is decreasing



Where is the CO2 coming from?

So we must have a lot of new CO2 being added to the atmosphere that has a low $^{14}C/^{12}C$ ratio (i.e. a source with no carbon-14)

What has this ratio?

Ancient plants/fossils!

C-14 decays over time, so only very old sources have non-detectable levels

Putting it together: Cause of rising CO2

FACT 1: CO2 concentration in the atmosphere is *increasing* (regardless of isotope)

FACT 2: The $^{13}C/^{12}C$ ratio is *decreasing*

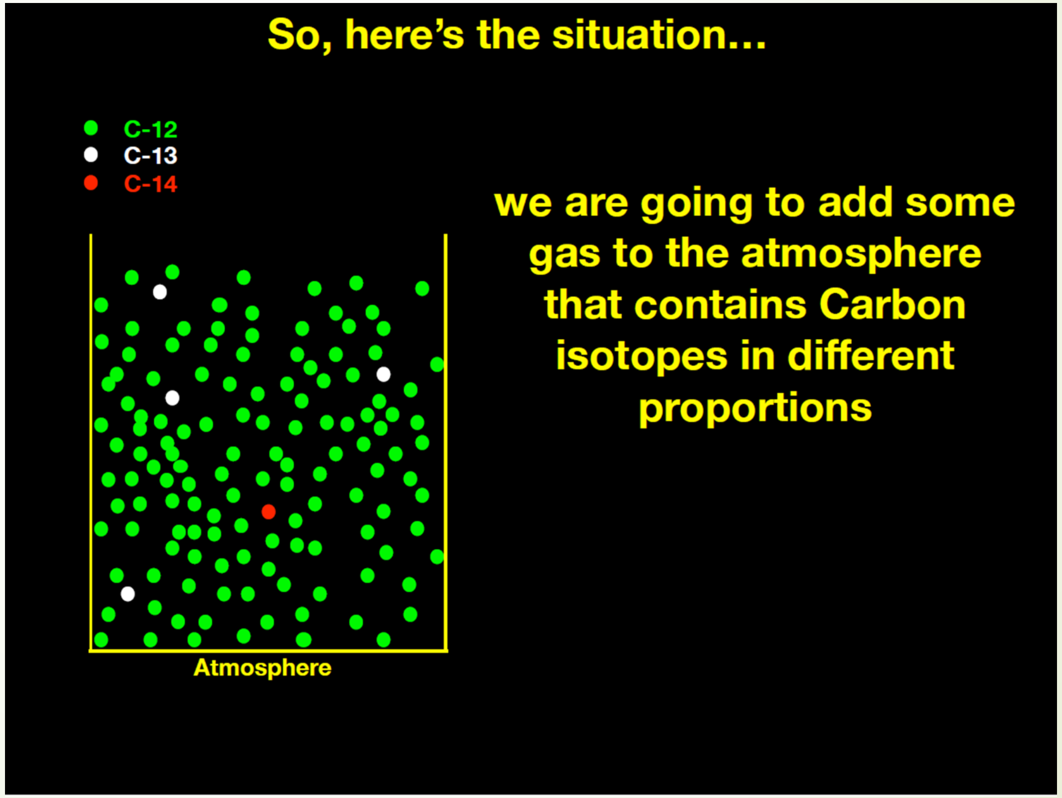
FACT 3: The $^{14}C/^{12}C$ ratio is *decreasing*

Only fossil fuels meet all those criteria.

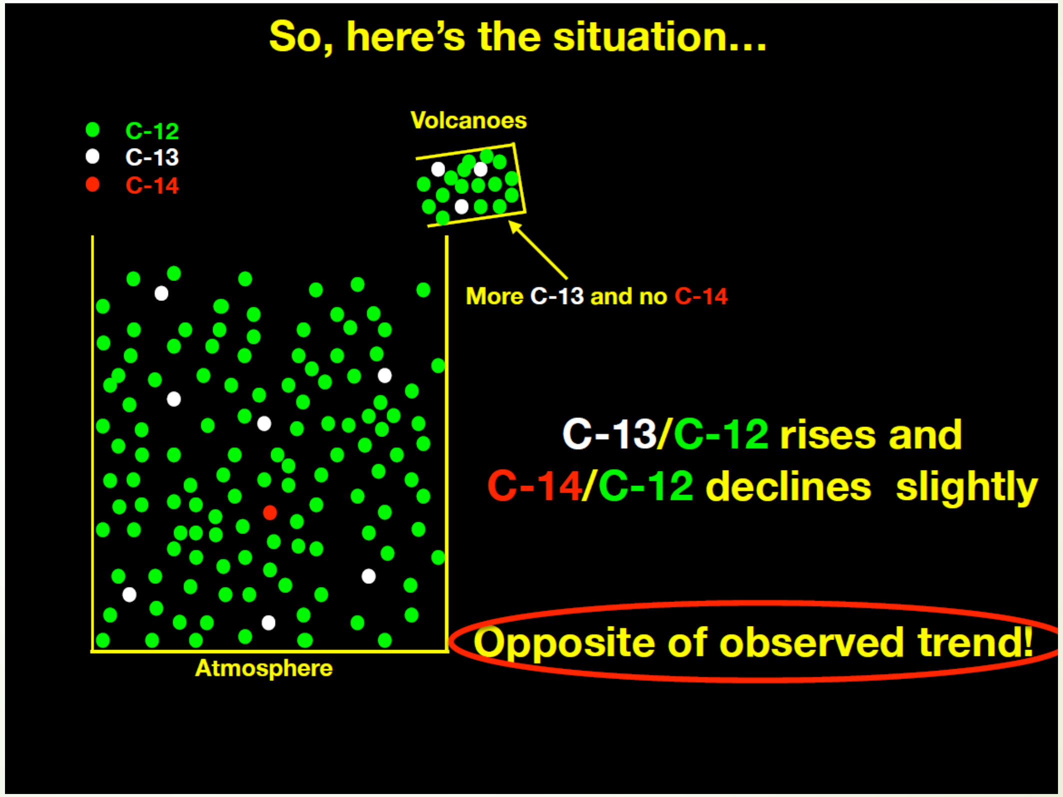
Isotopic evidence visualized

If you are having trouble visualizing how the RATIOS of 13C/12C and 14C/12C are both going down, despite adding CO2, here is a visualization from the lecture…

Isotopic evidence visualized



Isotopic evidence visualized



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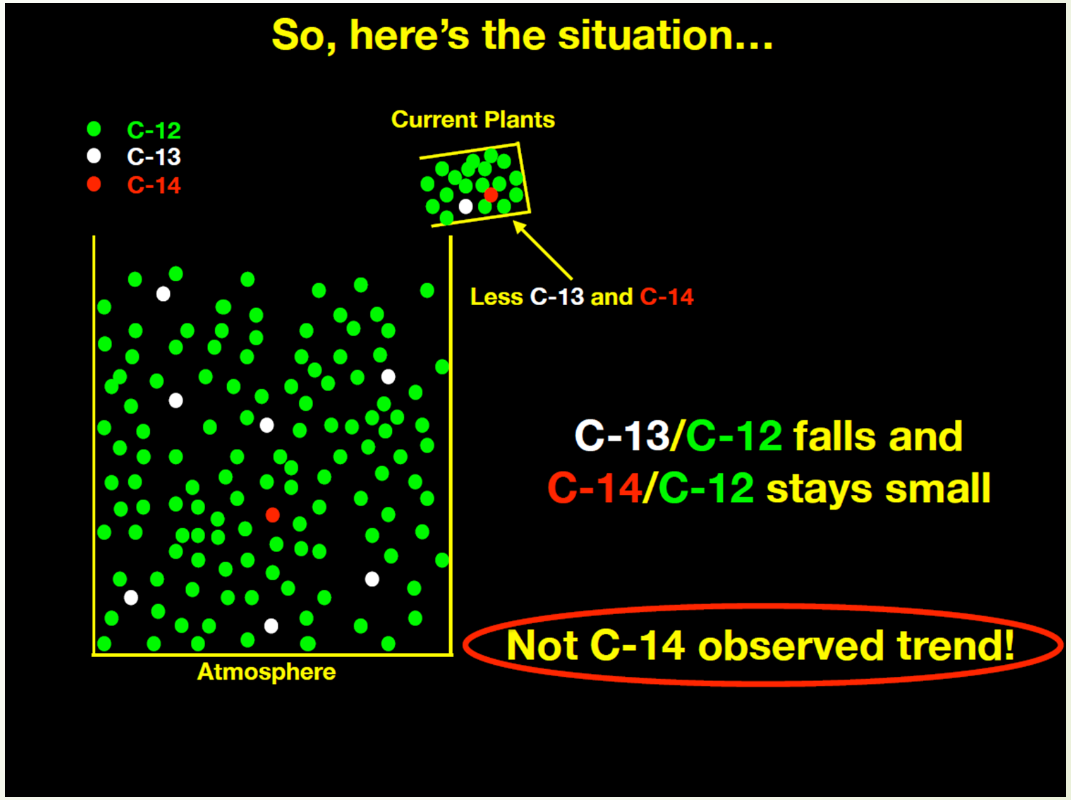
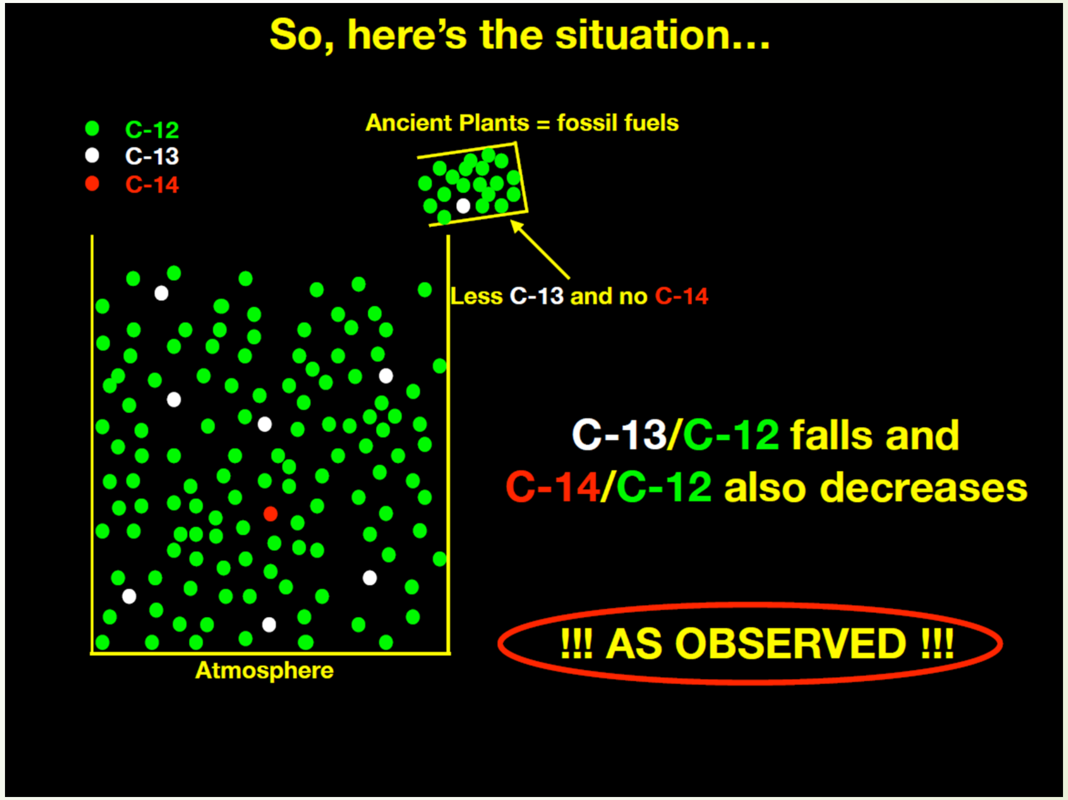
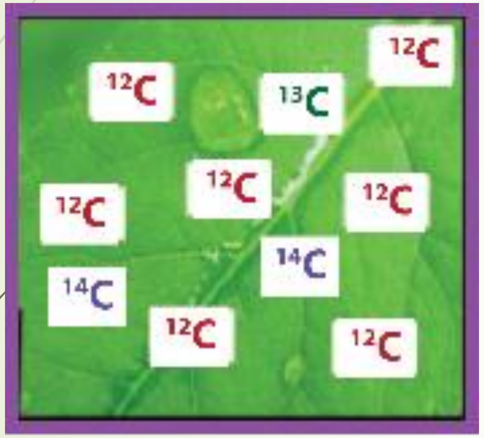


Isotopic evidence visualized



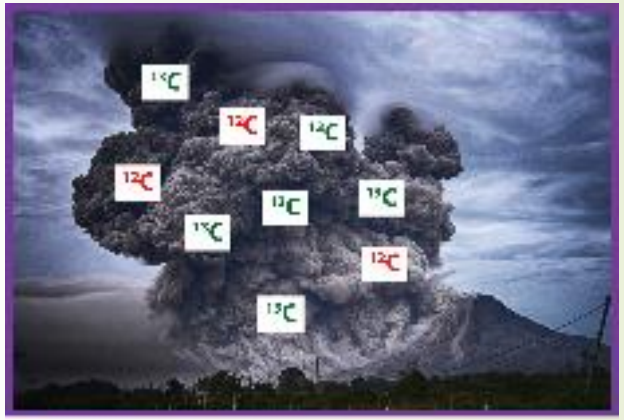
Part I: Isotopes Data

Low 13C/12C; Moderate 14C/12C



living vegetation

High 13C/12C; Zero to low 14C/12C



volcanoes

Low 13C/12C; Zero 14C/12C

fossil fuels

Part I: Isotopes Data

The 13C/12C ratio of the atmosphere is decreasing with time, showing that the carbon dioxide being added to the atmosphere must have a low 13C/12C ratio. Volcanoes have a high 13C/12C ratio, and CO_2 only from volcanoes would increase the 13C/12C ratio of the atmosphere. Two sources of carbon dioxide that have a low 13C/12C ratio are vegetation or fossil fuels. Thus, based on observed 13C/12C alone, the changes in CO2 isotopes could be a result of these, but cannot be due to volcanoes.

The 14C/12C ratio of the atmosphere has also been decreasing with time, showing that the carbon dioxide being added to the atmosphere must have a low 14C/12C ratio. Living vegetation has a higher 14C/12C ratio than the 14C/12C ratio of volcanoes or fossil fuels. Therefore, the decreasing 13C/12C and 14C/12C ratios in the atmosphere must be caused by adding large amounts of carbon dioxide from fossil fuels.

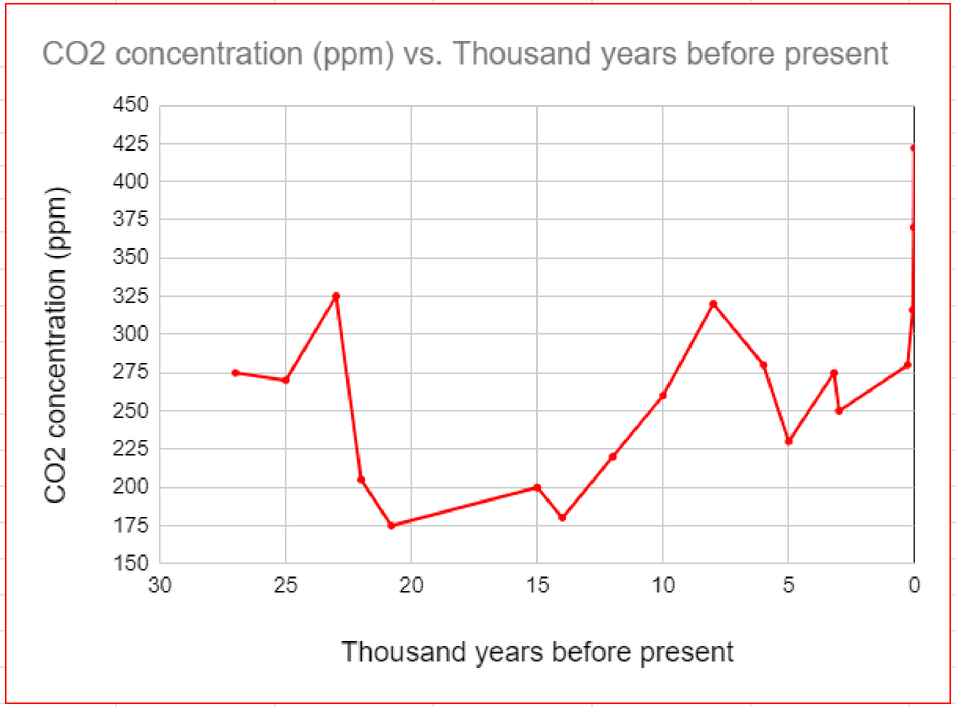
Part II: Ice Cores

ENTER ICE CORE DATA IN THE RED FIELDS:

ENTER MEASURED DATA IN BLUE FIELDS:

Measured:		Thousand years before present	CO2 concentration (ppm)
	2023	0	422
	2000	0.023	370
	1960	0.063	316
	1750	0.273	280
Ice Cores:		3	250
		3.2	275
		5	230
		6	280
		8	320
		10	260
		12	220
		14	180
		15	200
		20.8	175
		22	205
		23	325
		25	270
		27	275

Figure 4: Concentration of CO2 (ppm) vs. time from Antarctic ice cores



What is the maximum CO_2 concentration you found in the <u>air bubble</u> data? What is the most recent <u>measurement</u> of atmospheric CO_2 concentration?

Max: 325 ppm
2025: 425 ppm

Calculate the rate of change of CO_2 concentration (in ppm/year) from approximately 10,000 years ago to 8,000 years ago?.

0.03 ppm CO_2 per year

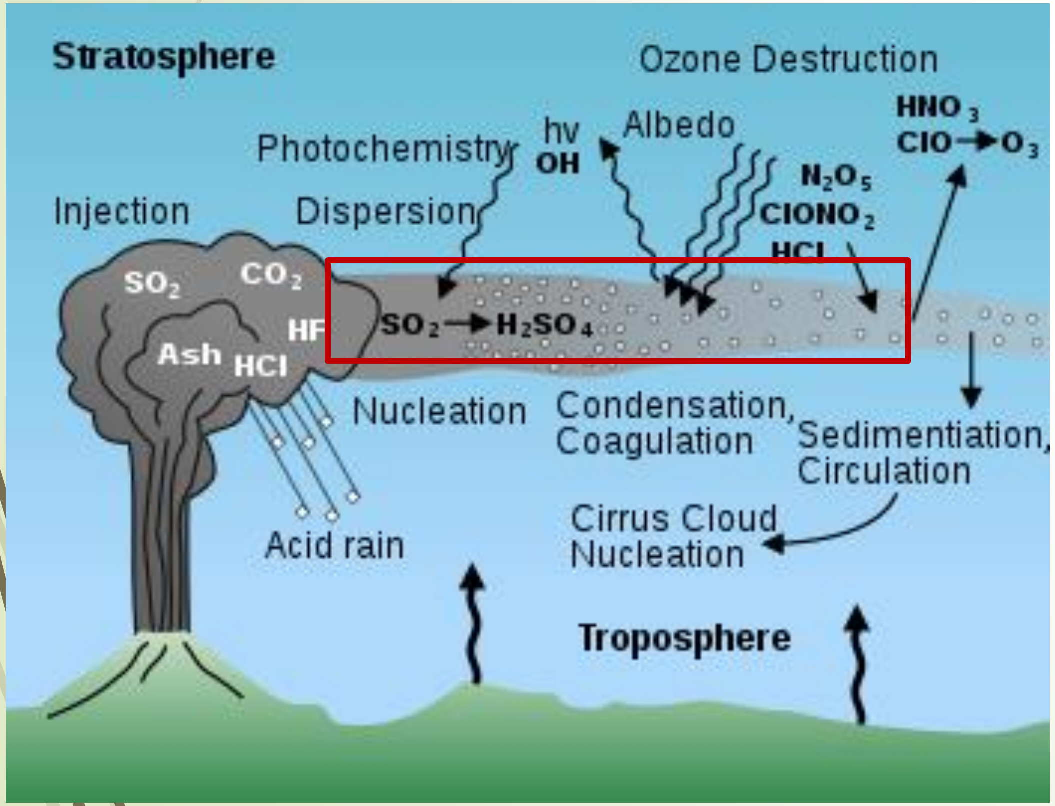
How many times larger is the rate of change from 2000-2021 in CO_2 concentration compared to the rate of change you calculated from 10,000 to 8,000 years ago?

~73 X larger

Part III: Volcanoes

What are two major ways that volcanoes can affect climate?

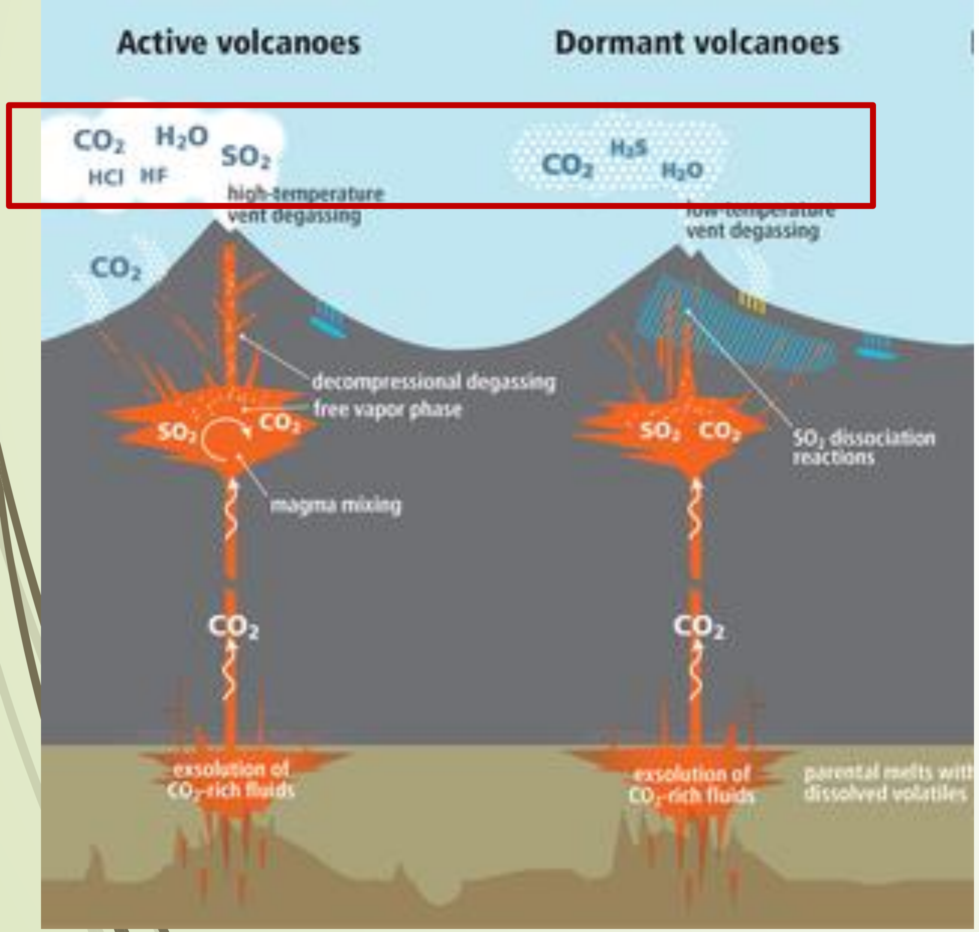
SO_2



- $SO2$ interacts with water vapor → converted to sulfate particles
- Sulfate particles increase albedo of volcanic cloud
- Volcanic cloud circulates globe, creating a cooling effect

Part III: Volcanoes

What are two major ways that volcanoes can affect climate?



CO_2

- Active volcanic eruptions emit CO2
- Dormant volcanoes also emit CO2 through passive degassing

Part III: Volcanoes

1991 Eruption of Mt. Pinatubo, Philippines

usgs.gov

How did it affect climate?

0.6°C cooled globally for three years

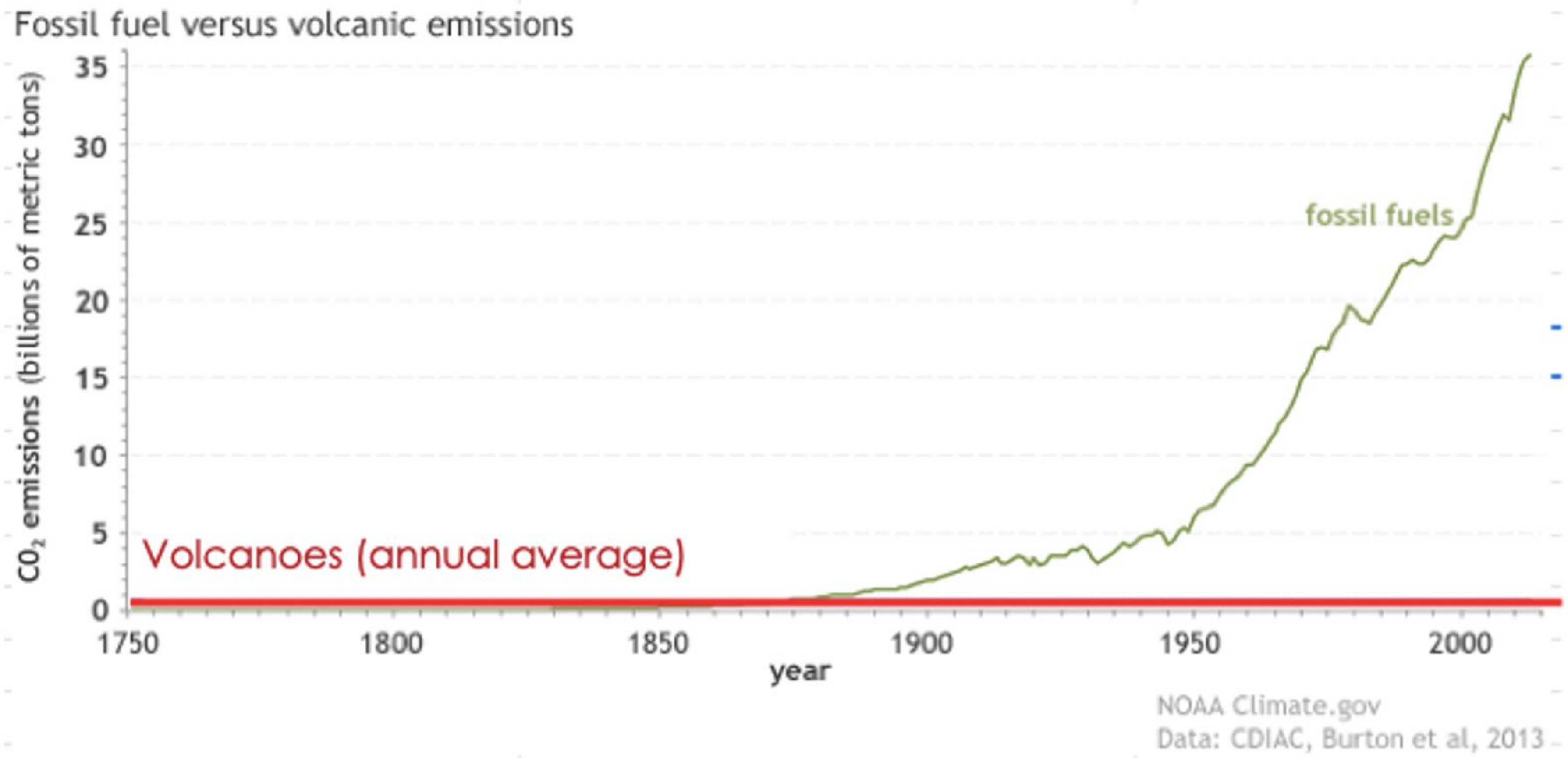
How much CO2 emitted?

0.054 gigatons

Part III: Volcanoes

How many Pinatubo eruptions would we need? Is that possible?

648 Pinatubo eruptions per year



Putting at all together

You now have multiple pieces of data (from isotopes, ice cores, and volcanoes) as evidence of the impacts of human activities on recent climate change.

Write a short elevator speech to your local congressperson to make the argument that human use of fossil fuels is the cause of the recent increase in atmospheric CO_2 concentration.

Be sure to use the data you've collected to support your argument. You may work in groups!

Rising CO2: What is the cause? Evidence *against* volcanoes:

1. Amount of CO_2 is increasing rapidly
   - Volcanoes have **not** been erupting more frequently
   - CO_2 emissions from humans is > 60 x emissions of all volcanoes globally
2. O_2 is also decreasing, perfectly anti-correlated with increasing CO_2
   - Volcanoes emit CO_2 produced by melting carbonate sediments (doesn't use O_2)
   - Burning fossil fuels requires O_2 to burn ancient plant material
3. $^{13}C/^{12}C$ and $^{14}C/^{12}C$ are both decreasing
   - Volcanic CO_2 has high ^{13}C: more emission wouldn't $\downarrow ^{13}C/^{12}C$
   - Only fossil fuels have both low $^{13}C/^{12}C$ and low $^{14}C/^{12}C$

Great 2-min Summary: https://www.youtube.com/watch?v=UXgDrr6qiUk&t=9s